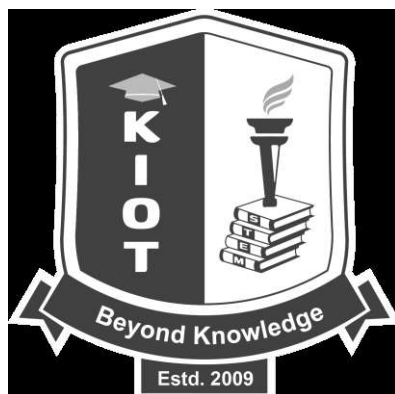


KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM

(An Autonomous Institution)

Approved by AICTE, Affiliated to Anna University, Chennai.
Accredited by NBA (CSE, ECE, EEE & MECH), Accredited by NAAC with 'A' Grade

KIOT Campus, Kakapalayam – 637 504. Salem Dt., Tamil Nadu, India.



M.E./M.Tech./M.C.A. Regulations 2023

MASTER OF COMPUTER APPLICATIONS

Curriculum and Syllabi

(For the Students Admitted from the Academic Year 2025 – 2026 onwards)

Version: 1.0	Date: 15.12.2025
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KNOWLEDGE INSTITUTE OF TECHNOLOGY(AUTONOMOUS), SALEM -637504

Approved by AICTE, Affiliated to Anna University,
Accredited by NAAC and NBA (B.E.: Mech., ECE, EEE & CSE)

Website: www.kiot.ac.in

M.C.A. REGULATIONS 2023 (R 2023)

M.C.A. – MASTER OF COMPUTER APPLICATIONS

CHOICE BASED CREDIT SYSTEM AND OUTCOME BASED EDUCATION

VISION OF THE INSTITUTE

To be a world class institution to impart value and need based professional education to the Aspiring youth and carving them into disciplined world class professional who have the quest for excellence, achievement orientation and social responsibilities.

MISSION OF THE INSTITUTE

A	To promote academic growth by offering state-of-art undergraduate, postgraduate and doctoral programs and to generate new knowledge by engaging in cutting – edge research
B	To nurture talent, innovation, entrepreneurship, all-round personality and value system among the students and to foster competitiveness among students
C	To undertake collaborative projects which offer opportunities for long-term interaction with academia and industry
D	To pursue global standards of excellence in all our endeavors namely teaching, research, consultancy, continuing education and support functions

VISION OF THE DEPARTMENT

To be a state-of-the-art Department of Computer Applications that produces competent professionals with social, ethical, and environmental values.

MISSION OF THE DEPARTMENT

A	Provide advanced postgraduate education in Computer Applications integrating emerging technologies and best industry practices
B	Cultivate professional ethics, leadership qualities and lifelong learning commitment.
C	Empower students for successful careers and entrepreneurship through comprehensive skill and personality development.
D	Foster innovation, research aptitude and problem-solving through academic-industry collaboration

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Knowledge Institute of Technology
KIOT Campus, Kakapalayam
Salem-637 504

w.e.f. 22.12.2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15.12.2025

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)	
PEO 1	Develop professional expertise in advanced computer applications and software development to solve complex computational and industrial problems.
PEO 2	Cultivate analytical thinking, research aptitude, and lifelong learning for leadership roles and higher education.
PEO 3	Encourage ethical practices, innovation, and entrepreneurship to design sustainable computing solutions addressing societal and industrial needs.

PROGRAM OUTCOMES (POs)	
MCA Graduates will be able to:	
PO1	Foundation Knowledge: Apply knowledge of mathematics, programming logic and coding fundamentals for ability to Computer Applications.
PO2	Problem Analysis: Identify, formulate and analysis problems based on customer requirements.
PO3	Development of Solutions: Design and develop efficient computer solutions and algorithms.
PO4	Modern Tool Usage: Select, adapt and apply modern computational tools for application development.
PO5	Project Management and Finance: Ability to understand and apply the concepts of Project Management and finance.
PO6	Communication and Presentation: Ability to make effective presentations and prepare reports and documents.
PO7	Individual and Teamwork: Ability to perform as an individual and as a members of a team.
PO8	Ethics: Commit to professional ethics, environment and sustainable practices in managing software projects.
PO9	Life-long learning: Ability to learn, keep up with contemporary technologies and ways of working continuously.

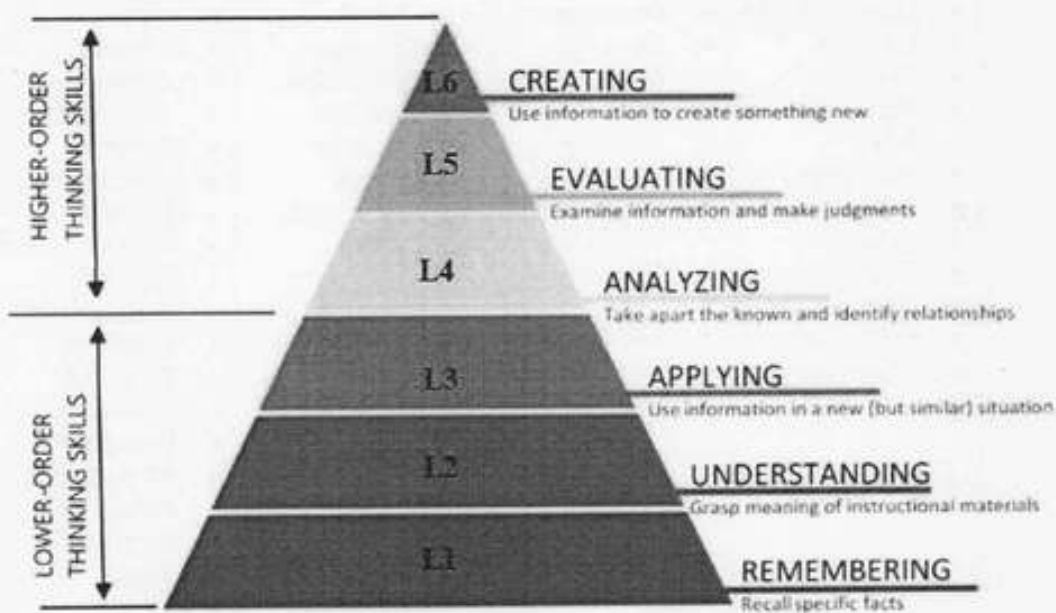

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BLOOM'S TAXONOMY LEVELS AND ACTION VERBS

(A) BLOOM'S TAXONOMY LEVELS (BTL)

Bloom's Taxonomy (BT) is based on the belief that learners must first acquire basic foundational knowledge about a subject before progressing to more complex types of thinking, such as analysis and evaluation. Bloom's Taxonomy levels help faculty to guide students through the learning process, from fundamental remembering and understanding to more complex evaluating and creating.



At KIOT, Curriculum Design, Delivery and Assessment (CDDA) are carried out based on the Blooms' Taxonomy Levels (BTL). Its organized set of objectives helps teachers to plan and deliver appropriate instruction, design valid assessment tasks and schemes, and ensure that instruction and assessment are aligned with the objectives.


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(B) BLOOM'S TAXONOMY ACTION VERBS

I. Remembering	II. Understanding	III. Applying	IV. Analyzing	V. Evaluating	VI. Creating
Exhibit memory of Previously Learned material by recalling facts, terms, basic concepts, and answers.	Demonstrate understanding of facts and ideas by organizing, comparing, interpreting, giving descriptions, and stating main ideas.	Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations.	Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria.	Compile information together in a different way by combining elements in a new pattern or proposing new Solutions.
- Choose - Define - Find - How - Label - List - Match - Name - Omit - Recall - Relate - Select - Show - Spell - Tell - What - When - Where - Which - Who - Why	- Classify - Compare - Contrast - Demonstrate - Explain - Extend - Illustrate - Infer - Interpret - Outline - Relate - Rephrase - Show - Summarize - Translate	- Apply - Build - Choose - Construct - Develop - Experiment with - Identify - Interview - Make use of - Model - Organize - Plan - Select - Solve - Utilize	- Analyze - Assume - Categorize - Classify - Compare - Conclusion - Contrast - Discover - Dissect - Distinguish - Divide - Examine - Function - Inference - Inspect - List - Motive - Relationships - Simplify - Survey - Take part in - Test for - Theme	- Agree - Appraise - Assess - Award - Choose - Compare - Conclude - Criteria - Criticize - Decide - Deduct - Defend - Determine - Disprove - Estimate - Evaluate - Explain - Importance - Influence - Interpret - Judge - Justify - Mark - Measure - Opinion - Perceive - Prioritize - Prove - Rate - Recommend - Rule on - Select - Support - Value	- Adapt - Build - Change - Choose - Combine - Compile - Compose - Construct - Create - Delete - Design - Develop - Discuss - Elaborate - Estimate - Formulate - Happen - Imagine - Improve - Invent - Make up - Maximize - Minimize - Modify - Original - Originate - Plan - Predict - Propose - Solution - Solve - Suppose - Test - Theory

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KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM – 637504												
MASTER OF COMPUTER APPLICATIONS (2 YEARS)										Version : 1.0		
Courses of Study and Scheme of Assessment (Regulations 2023)										Date:15.12.2025		
S. No.	Course Code	Course Title	Periods / Week						Maximum Marks			
			CAT	CP	L	T	P	C	IA	ESE	Total	
SEMESTER I												
1	MC23CA901	Problem Solving and Programming in C	BC	5	3	0	2	NC	50	50	100	
THEORY												
1	MC23MA101	Mathematical Foundation for Computer Applications	FC	4	3	1	0	4	40	60	100	
2	UX23RM201	Research Methodology and IPR	RM	3	2	1	0	3	40	60	100	
3	MC23CA301	Advanced Computer Networks	PC	3	3	0	0	3	40	60	100	
4	MC23CA302	Data Structure and Algorithms using C	PC	3	3	0	0	3	40	60	100	
5	MC23CA303	Advanced Database Technology	PC	3	3	0	0	3	40	60	100	
6	MC23CA304	Object Oriented Programming using Java	PC	3	3	0	0	3	40	60	100	
PRACTICAL												
7	MC23CA305	Data Structures and Algorithms Laboratory	PC	4	0	0	4	2	60	40	100	
8	MC23CA306	Advanced Database Technology Laboratory	PC	4	0	0	4	2	60	40	100	
9	MC23CA307	Object Oriented Programming using Java Laboratory	PC	4	0	0	4	2	60	40	100	
10	MC23CA308	Fundamentals of Linux and Scripting Laboratory	PC	2	0	0	2	1	60	40	100	
EMPLOYABILITY ENHANCEMENT												
11	MC23PT701	Communication Skills Enhancement - I	AC	2	0	0	2	NC	100	-	100	
Total				35	17	2	16	26	580	520	1100	
SEMESTER II												
THEORY												
1	MC23CA309	Cybersecurity	PC	3	3	0	0	3	40	60	100	
2	MC23CA4XX	Professional Elective - I	PE	3	3	0	0	3	40	60	100	
3	MC23CA4XX	Professional Elective - II	PE	3	3	0	0	3	40	60	100	
4	MC23XX5XX	Open Elective - I	OE	3	3	0	0	3	40	60	100	
5	UX23MC701	Universal Human Values and Ethics	MC	3	1	0	2	NC	100	-	100	
THEORY CUM PRACTICAL												
6	MC23CA310	Internet of Things	PC	4	2	0	2	3	50	50	100	
PRACTICAL												
7	MC23CA4XX	Professional Elective - I Laboratory	PE	4	0	0	4	2	60	40	100	
8	MC23CA4XX	Professional Elective - II Laboratory	PE	4	0	0	4	2	60	40	100	
9	MC23CA601	Capstone Project - I	PW	4	0	0	4	2	60	40	100	
EMPLOYABILITY ENHANCEMENT												
10	MC23PT702	Communication Skills Enhancement - II	AC	2	0	0	2	NC	100	-	100	
11	MC23PT801	Aptitude Skills - I	EEC	2	0	0	2	1	100	-	100	
12	UX23PT802	Research Paper Review and Presentation	EEC	2	0	0	2	1	100	-	100	
Total				15	0	22	23	790	410	1200		

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MASTER OF COMPUTER APPLICATIONS (2 YEARS)											
Courses of Study and Scheme of Assessment (Regulations 2023)											
S. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
SEMESTER III											
THEORY											
1	MC23CA311	Big Data Analytics	PC	3	3	0	0	3	40	60	100
2	MC23CA4XX	Professional Elective - III	PE	3	3	0	0	3	40	60	100
3	MC23CA4XX	Professional Elective - IV	PE	3	3	0	0	3	40	60	100
4	MC23XX5XX	Open Elective - II	OE	3	3	0	0	3	40	60	100
THEORY CUM PRACTICAL											
5	MC23CA312	Machine Learning	PC	4	2	0	2	3	50	50	100
PRACTICAL											
6	MC23CA4XX	Professional Elective - III Laboratory	PE	4	0	0	4	2	60	40	100
7	MC23CA4XX	Professional Elective – IV Laboratory	PE	4	0	0	4	2	60	40	100
8	MC23CA602	Capstone Project - II	PW	4	0	0	4	2	60	40	100
EMPLOYABILITY ENHANCEMENT											
9	MC23PT803	Soft Skills	EEC	2	0	0	2	NC	100	-	100
10	MC23PT804	Aptitude Skills - II	EEC	2	0	0	2	1	100	-	100
Total				32	14	0	18	22	590	410	1000
SEMESTER IV											
PRACTICAL											
1	MC23CA603	Project Work / Internship	PW	24	0	0	24	12	60	40	100
Total				24	0	0	24	12	60	40	100
Total Number of Credits: 83											

AUDIT COURSES / MANDATORY COURSE											
S. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1.	MC23PT701	Communication Skills Enhancement - I	AC	2	0	0	2	NC	100	-	100
2.	MC23PT702	Communication Skills Enhancement - II	AC	2	0	0	2	NC	100	-	100
MANDATORY COURSES											
1.	UX23MC701	Universal Human Values and Ethics	MC	3	1	0	2	NC	100	-	100


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SEMESTER-WISE CREDITS DISTRIBUTION & NOMENCLATURE

SUMMARY							
S. No.	Course Category	Credits per Semester				Credits	Credit %
		I	II	III	IV		
	BC	√	-	-	-	-	-
1.	FC	4	-	-	-	4	5
2.	PC	19	6	6	-	31	37
3.	PE	-	10	10	-	20	24
4.	PW	-	2	2	12	16	19
5.	EEC	-	2	1		3	4
6.	MC/AC/NC	√	√	√	-	-	-
7.	RM	3	-	-	-	3	4
8.	OE	-	3	3		6	7
Total		26	23	22	12	83	100

AT	Category of Course	C	Credits	OE	Open Elective Courses
CP	Contact Period	BC	Bridge Course	PW	Project Work Courses
L	Lecture Period	FC	Foundation Course	EEC	Employability Enhancement Course
T	Tutorial Period	PC	Professional Core Courses	MC	Mandatory Courses
P	Laboratory Period	PE	Professional Elective Courses	AC	Audit Course
RM	Research Methodology and IPR Courses				


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PROFESSIONAL ELECTIVE COURSES: VERTICALS

S. No.	Vertical I (Java Full - Stack Development)	Vertical II (Automation Testing)	Vertical III (General Vertical)
1	Web Development Fundamentals	Testing Fundamentals and Manual Testing	Exploratory Data Analytics and Visualization
2	Front - End Development	Automation Testing with Selenium	Cloud Services Management
3	Back- End Development with Java	Advanced Testing with TestNG and CI/CD	Blockchain Technology
4	Full - Stack Development and Deployment	Tools for Automated Testing	Large Language Models and AI agents
5	Database Management and Integration	Performance and Database Testing	Agile based Project Automation with DevOps
6	DevOps and Cloud Basics	Testing using AI and ML Tools	IT World Essentials
7	Web Development Fundamentals Laboratory	Testing Fundamentals and Manual Testing Laboratory	Exploratory Data Analytics and Visualization Laboratory
8	Front – End Development Laboratory	Automation Testing with Selenium Laboratory	Cloud Services Management Laboratory
9	Back-End Development with Java Laboratory	Advanced Testing with TestNG and CI/CD Laboratory	Blockchain Technology Laboratory
10	Full - Stack Development and Deployment Laboratory	Tools for Automated Testing Laboratory	Large Language Models and AI agents Laboratory
11	Database Management and Integration Laboratory	Performance and Database Testing Laboratory	Agile based Project Automation with DevOps Laboratory
12	DevOps and Cloud Basics Laboratory	Testing using AI and ML Tools Laboratory	IT World Essentials Laboratory


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VERTICAL I: JAVA FULL - STACK DEVELOPMENT

Sl. No	Course Code	Course Title	Periods / Week						Maximum marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1	MC23CA401	Web Development Fundamentals	PE	3	3	0	0	3	40	60	100
2	MC23CA402	Front-End Development	PE	3	3	0	0	3	40	60	100
3	MC23CA403	Back-End Development with Java	PE	3	3	0	0	3	40	60	100
4	MC23CA404	Full-Stack Development and Deployment	PE	3	3	0	0	3	40	60	100
5	MC23CA405	Database Management and Integration	PE	3	3	0	0	3	40	60	100
6	MC23CA406	DevOps and Cloud Basics	PE	3	3	0	0	3	40	60	100
7	MC23CA407	Web Development Fundamentals Laboratory	PE	4	0	0	4	2	60	40	100
8	MC23CA408	Front-End Development Laboratory	PE	4	0	0	4	2	60	40	100
9	MC23CA409	Back-End Development with Java Laboratory	PE	4	0	0	4	2	60	40	100
10	MC23CA410	Full-Stack Development and Deployment Laboratory	PE	4	0	0	4	2	60	40	100
11	MC23CA411	Database Management and Integration Laboratory	PE	4	0	0	4	2	60	40	100
12	MC23CA412	DevOps and Cloud Basics Laboratory	PE	4	0	0	4	2	60	40	100

VERTICAL II: AUTOMATION TESTING

Sl. No	Course Code	Course Title	Periods / Week						Maximum marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1	MC23CA413	Testing Fundamentals and Manual Testing	PE	3	3	0	0	3	40	60	100
2	MC23CA414	Automation Testing with Selenium	PE	3	3	0	0	3	40	60	100
3	MC23CA415	Advanced Testing with TestNG and CI/CD	PE	3	3	0	0	3	40	60	100
4	MC23CA416	Tools for Automated Testing	PE	3	3	0	0	3	40	60	100
5	MC23CA417	Performance and Database Testing	PE	3	3	0	0	3	40	60	100
6	MC23CA418	Testing using AI and ML Tools	PE	3	3	0	0	3	40	60	100

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7	MC23CA419	Testing Fundamentals and Manual Testing Laboratory	PE	4	0	0	4	2	60	40	100
8	MC23CA420	Automation Testing with Selenium Laboratory	PE	4	0	0	4	2	60	40	100
9	MC23CA421	Advanced Testing with TestNG and CI/CD Laboratory	PE	4	0	0	4	2	60	40	100
10	MC23CA422	Tools for Automated Testing Laboratory	PE	4	0	0	4	2	60	40	100
11	MC23CA423	Performance and Database Testing Laboratory	PE	4	0	0	4	2	60	40	100
12	MC23CA424	Testing using AI and ML Tools Laboratory	PE	4	0	0	4	2	60	40	100

VERTICAL III: GENERAL VERTICAL

Sl. No	Course Code	Course Title	Periods / Week						Maximum marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1	MC23CA425	Exploratory Data Analytics and Visualization	PE	3	3	0	0	3	40	60	100
2	MC23CA426	Cloud Services Management	PE	3	3	0	0	3	40	60	100
3	MC23CA427	Blockchain Technology	PE	3	3	0	0	3	40	60	100
4	MC23CA428	Large Language Models and AI agents	PE	3	3	0	0	3	40	60	100
5	MC23CA429	Agile based Project Automation with DevOps	PE	3	3	0	0	3	40	60	100
6	MC23CA430	IT World Essentials	PE	3	3	0	0	3	40	60	100
7	MC23CA431	Exploratory Data Analytics and Visualization Laboratory	PE	4	0	0	4	2	60	40	100
8	MC23CA432	Cloud Services Management Laboratory	PE	4	0	0	4	2	60	40	100
9	MC23CA433	Blockchain Technology Laboratory	PE	4	0	0	4	2	60	40	100
10	MC23CA434	Large Language Models and AI agents Laboratory	PE	4	0	0	4	2	60	40	100
11	MC23CA435	Agile based Project Automation with DevOps Laboratory	PE	4	0	0	4	2	60	40	100
12	MC23CA436	IT World Essentials Laboratory	PE	4	0	0	4	2	60	40	100


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OPEN ELECTIVES

Students will choose the open elective courses from the list of open electives such that the course contents are not similar to any other course contents/title under other course categories.

OPEN ELECTIVE-I (2nd Semester)

S. No.	Course Code	Course Title	Offered by	Periods / Week						Maximum marks		
				CAT	CP	L	T	P	C	IA	ESE	Total
1	MC23MB501	Principles of Management	MBA	OE	3	3	0	0	3	40	60	100
2	MC23MB502	Organizational Behavior	MBA	OE	3	3	0	0	3	40	60	100
3	MC23MB503	Fundamentals of Marketing	MBA	OE	3	3	0	0	3	40	60	100
4	MC23MB504	Accounting for Managers	MBA	OE	3	3	0	0	3	40	60	100

OPEN ELECTIVE-II (3rd Semester)

S. No	Course Code	Course Title	Offered by	Periods / Week						Maximum marks		
				CAT	CP	L	T	P	C	IA	ESE	Total
1	MC23MB505	Human Resource Management	MBA	OE	3	3	0	0	3	40	60	100
2	MC23MB506	Project Management	MBA	OE	3	3	0	0	3	40	60	100
3	MC23MB507	Supply Chain Management	MBA	OE	3	3	0	0	3	40	60	100
4	MC23MB508	Statistics for Business Analytics	S & H	OE	3	3	0	0	3	40	60	100
5	MC23MB509	Numerical Methods	S & H	OE	3	3	0	0	3	40	60	100


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SEMESTER – I

w.e.f. 13/10/2025
Passed in BoS Meeting held on 23/09/2025
Approved in Academic Council Meeting held on 11/10/2025

MC23CA901	PROBLEM SOLVING AND PROGRAMMING IN C	CP	L	T	P	C
		5	3	0	2	NC
Programme & Branch	M.C.A. - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To introduce students to the fundamentals of computational thinking and structured problem-solving techniques.					
2.	To implement these solutions efficiently using the C programming language					
3.	To emphasize structured programming principles, and the use of arrays, functions, and pointers for writing optimized code.					
4.	To accentuate structured programming principles, and the use of arrays, functions, and pointers for writing optimized code.					
5.	To gain hands-on experience in memory management and file handling in C.					
	INTRODUCTION (Not for Examination)					2
	Importance: Problem Solving and Programming in C is crucial as it builds a strong foundation in logical thinking, algorithm design, and coding efficiency. It enables students to develop optimized solutions for real-world computational problems. Real-time example: An embedded system in a washing machine uses C programming to control operations like water level detection, motor speed, and timer functions. Pre-requisite (s): • Basic Computer Knowledge • Logical and Analytical Thinking Fundamental Understanding of Programming Concepts					
UNIT-I	PROGRAMMING FUNDAMENTALS					9
	Computational Thinking - Programming Concepts - Software - Software Development Life Cycle - Problem Solving using Algorithm - Flowchart (Using Flowgorithm) - Pseudocode.					
UNIT-II	INTRODUCTION TO C PROGRAMMING					9
	Introduction to C- Features - Structure of C Programming - Environmental Setup (Compilation and Execution) - Character Set - Tokens (Keywords, Identifiers, Constants, Variables, Operators) - Data Types - Expressions - Input /Output Operations - Type Conversion and Type Casting- Control Flow Statements: Sequence, Selection, and Looping.					
UNIT- III	ARRAYS, STRINGS, AND POINTER BASICS					9
	Arrays: Introduction - Types - Single Dimensional (Declaration, Initialization, Accessing) - Multi-Dimensional Arrays. Strings: Introduction - Inbuilt Functions - Two-Dimensional Character Arrays. Pointers: Introduction - Pointer to Pointer - Pointer Arithmetic - Pointer Types (NULL, void, Wild, Near, Far, and Huge Pointers) - Pointer and Arrays - Pointer and Strings - Array of Pointers .					

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UNIT – IV	FUNCTIONS AND ADVANCED POINTERS	9
	Functions: Introduction – Types (Library / Built-in, User-defined) – Components of User-defined Functions – Actual Arguments and Formal Parameters – Pass by Value and Pass by Reference – Recursive Functions – Storage Classes – Scope and Lifetime of Variables. Advanced Pointers: Pointer to an Array – Function Pointer – Memory Management – Static and Dynamic Memory Allocation – Dangling Pointers – Memory Leak – Type Qualifiers .	
UNIT-V	STRUCTURE, FILE MANAGEMENT, AND OTHER FEATURES	9
	Structure: Introduction – Declaration – Initialization – Accessing – Assignment – Nested Structure– Array of Structures – Pointer to Structure – Structure Padding – Self Referential Structure – Structure and Functions – Unions and Bit Fields. File Management: Introduction – File Handling – Types – Input and Output Functions – Random File Access. Other Features: Introduction to Pre-processor Directives – Types – Command Line Arguments. .	
Total		47 Periods
LIST OF EXPERIMENTS		
1. Design Algorithm, flowcharts and pseudocode for real-world problem scenarios. 2. Write and execute basic C programs covering variables, data types, and operators. 3. Implement all control flow statements with simple programs and real-world examples. 4. Implement arrays and string manipulation programs. 5. Demonstrate pointer concepts using simple programs. 6. Implement user-defined functions with different parameter passing techniques. 7. Develop memory management programs using malloc(), calloc(), free(). 8. Implement structure-based programs. 9. Perform file operations including reading, writing, and modifying files.		
Total		30 Periods
Total (L+P)		77 Periods
OPEN-ENDED PROBLEMS / QUESTIONS		
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy
CO1	Understand computational thinking and problem-solving techniques.	L2 – Understand
CO2	Design algorithms and represent those using flowcharts and pseudocode.	L2 – Understand
CO3	Develop C programs using structured programming principles.	L3 – Apply
CO4	Apply concepts of arrays, pointers, and functions to write efficient programs.	L3 – Apply
CO5	Implement memory management techniques and file handling operations in C.	L3 – Apply

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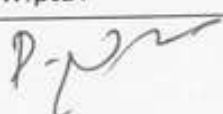
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TEXTBOOKS				
1.	E. Balagurusamy, "Programming in ANSI C", Seventh Edition, McGraw Hill, 2020			
2.	Yashavant Kanetkar, "Let Us C", Fifteenth Edition, BPB Publications, 2022.			
REFERENCE BOOKS				
1.	Brian W. Kernighan and Dennis M. Ritchie, "The C Programming Language", Second Edition, Prentice Hall, 1988.			
2.	Paul Deitel, Harvey Deitel, "C How to Program", Ninth Edition, Pearson Education, 2020.			
3.	Herbert Schildt, "C: The Complete Reference", Fourth Edition, McGraw Hill, 2000.			
WEB REFERENCES				
S. No.	Publisher	Website link		Type of Content
1.	Geeksforgeeks	https://www.geeksforgeeks.org/c-programming-language/		Web References
2.	Tutorialspoint	https://www.tutorialspoint.com/cprogramming/index.htm		Web References
3.	Scratch	https://scratch.mit.edu/		Web References
VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Jennys	Lecture	https://www.youtube.com/playlist?list=PLKh-P_-rjZjQkyYmfOToBIe8Ee4wPHbJT
2.	NPTEL	Techmania	Lecture	https://www.youtube.com/playlist?list=PLKh-P_-rjZjQkyYmfOToBIe8Ee4wPHbJT
3.	YouTube	Yashavant Kanetkar	Lecture	https://www.youtube.com/watch?v=AV7hmWfptdY


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Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
C01	3	3	2						2
C02	3	3	3						2
C03	3	2	3	3					1
C04	3	2	3	3					1
C05	3	2	3	3					1
Average	3	2.4	2.8	3					1.4
1-Low, 2-Medium, 3-High									



P. N. 2

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MC23MA101	MATHEMATICAL FOUNDATION FOR COMPUTER APPLICATIONS	CP	L	T	P	C
		4	3	1	0	4
Programme & Branch	MASTER OF COMPUTER APPLICATIONS	Version: 1.0				
Course Objectives:						
1.	To apply logical reasoning using propositional and predicate calculus.					
2.	To model and solve optimization problems using linear programming.					
3.	To apply PERT and CPM techniques for project scheduling and management.					
4.	To use numerical methods to solve equations and perform interpolation.					
5.	To strengthen analytical thinking to apply mathematics in computer science applications.					
	INTRODUCTION (Not for Examination)					2
Importance	This course builds a strong mathematical foundation essential for understanding logic, algorithms, optimization, and computational techniques. It enables students to analyze, model, and solve real-world problems in computer science, software development, and data-driven applications.					
Real-life Example(s)	Elevator Control System: Uses logic conditions to decide when to move up/down or open/close doors based on button inputs. Smart City Infrastructure Projects: Used by metro rail corporations (e.g., Chennai Metro) to plan construction timelines and avoid project delays. Google Maps ETA Prediction: Newton-Raphson method is used in route optimization and solving nonlinear models for real-time traffic prediction.					
Linkages	-					
UNIT-I	PROPOSITIONAL CALCULUS					9+3
	Propositions - Logical connectives - Compound propositions - Conditional and biconditional propositions - Truth tables Tautologies and contradictions - Contra positive - Logical equivalences and implications - Normal forms - Principal conjunctive and disjunctive normal forms .					
UNIT-II	PREDICATE CALCULUS					9+3
	Predicates - Statement functions - Variables - Free and bound variables - Quantifiers - Universe of discourse - Logical equivalences and implications for quantified statements - Theory of Inference - The rules of universal specification and generalization.					
UNIT-III	LINEAR PROGRAMMING					9+3
	Formation of Linear Programming Problems - Graphical method - Simplex method - Big M Method.					
UNIT - IV	NETWORK ANALYSIS -PERT/CPM					9+3
	Network Construction - Critical Path Method - Computation of earliest start time, latest start time, Total, free and independent float time - PERT -					

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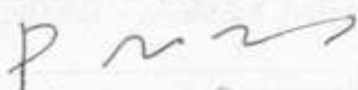
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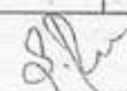

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	Computation of optimistic, most likely Pessimistic and expected time - Probabilistic estimation for completion of project.		
UNIT-V	NUMERICAL ANALYSIS	9+3	
	Solution of algebraic and transcendental equation – Newton Raphson's method – Regula falsi method – Interpolation and approximation – Newton's forward and backward Interpolation method.		
	Total (LT)	62 Periods	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy	
CO1	Apply rules of logic to analyze and validate arguments.	L3 – Apply	
CO2	Use predicates and quantifiers to represent real-world problems.	L3 – Apply	
CO3	Formulate and solve linear programming problems using appropriate methods.	L3 – Apply	
CO4	Apply PERT/CPM techniques to plan and analyze project activities.	L3 – Apply	
CO5	Use numerical methods to approximate solutions of equations and interpolate values.	L3 – Apply	
	REFERENCE BOOKS:		
1.	Kenneth H. Rosen "Discrete Mathematics and Its Applications", McGraw-Hill Edition, 8th edition, 2012.		
2.	Hamdy A. Taha, "Operations Research: An Introduction", Pearson, 10th Edition, 2017.		
3.	Richard L. Burden and J. Douglas Faires, "Numerical Analysis, Cengage Learning", 10th Edition, 2015.		
4.	Eric Lehman, F.Thomson Leighton, Albert R. Meyer, "Mathematics for Computer Science", MIT Press, 7th Edition, 2015.		
5.	P.K.Gupta & Man Mohan,"Operations Research", Sultan Chand & Sons Publications, 12th Edition, 2013.		
6.	Dr. M.K. Venkataraman "Numerical Methods in Science and Engineering", The National Publishing Company, 5th edition, 2016.		
	WEB REFERENCES		
S.No.	Publisher	Website link	Type of Content
1.	Tutorialspoint	https://www.tutorialspoint.com/discrete_mathematics/discrete_mathematics_propositional_log ic.htm	Lecture Notes
2.	Geeks for Geeks	https://www.geeksforgeeks.org/maths/linear-programming/	Lecture Notes


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	Mapping of COs with POs								
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2		1					
CO2	3	2		1					
CO3	3	2							
CO4	3	2			3				
CO5	3	2							
Avg.	3.0	2.0		1.0	3.0				
	1-Low, 2 -Medium, 3-High.								

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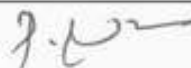
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UX23RM201	RESEARCH METHODOLOGY AND IPR	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	Common to all M.E (SE,AE,PED,ISE) and M.C.A	Version: 1.0				
Course Objectives:						
1.	To analyze the significance of research and formulate well-defined research questions.					
2.	To apply appropriate research methods and critically evaluate research articles.					
3.	To create well-structured research papers and utilize research tools proficiently.					
4.	To produce effective technical reports and deliver impactful presentations.					
5.	To understand forms of intellectual property and analyze their implications on technological research and international cooperation.					
UNIT-I	CONCEPT OF RESEARCH					6+3
	Meaning and Significance of Research-Skills, Habits and Attitudes for Research -Time Management-Status of Research in India -Why, How, and What a Research is? -Types and Process of Research -Outcome of Research -Sources of Research Problem -Characteristics of a Good Research Problem -Errors in Selecting a Research Problem -Importance of Keywords -Literature Collection - Analysis -Citation Study - Gap Analysis -Problem Formulation Techniques .					
UNIT-II	RESEARCH METHODS AND JOURNALS					6+3
	Interdisciplinary Research -Need for Experimental Investigations -Data Collection Methods -Appropriate Choice of Algorithms / Methodologies / Methods -Measurement and Result Analysis -Investigation of Solutions for Research Problem -Interpretation -Research Limitations - Journals in Science/Engineering -Indexing and Impact factor of Journals -Citations- h Index - i10 Index -Journal Policies How to Read a Published Paper -Ethical Issues Related to Publishing- Plagiarism and Self-Plagiarism .					
UNIT-III	PAPER WRITING AND RESEARCH TOOLS					6+3
	Types of Research Papers - Original Article/Review Paper/Short Communication/Case Study-When and Where to Publish? - Journal Selection Methods -Layout of a Research Paper -Guidelines for Submitting the Research Paper -Review Process - Addressing Reviewer Comments -Use of tools / Techniques for Research -Hands-on Training related to Reference Management Software - EndNote - Introduction to Origin, SPSS,etc -Software for Detection of Plagiarism					
UNIT - IV	EFFECTIVE TECHNICAL THESIS WRITING/PRESENTATION					6+3
	How to Write a Report- - Language and Style -Format of Project Report - Use of Quotations -Method of Transcription Special Elements - Title Page- - Abstract - Table of Contents - Headings and Sub-Headings - Footnotes - Tables and Figures - Appendix- Bibliography etc. -Different Reference Formats -Presentation using PPTs .					



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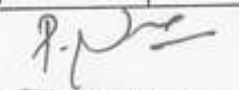
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UNIT – V	NATURE OF INTELLECTUAL PROPERTY	6+3	
	Patents - Designs - Trade and Copyright - Process of Patenting and Development - Technological research- innovation- - patenting- Development International Scenario -International Cooperation on Intellectual Property -Procedure for Grants of Patents .		
	Total (LT)	45 Periods	
	OPEN-ENDED PROBLEMS / QUESTIONS		
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.		
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy	
CO1	Illustrate the importance and objectives of research in contributing to knowledge and solving real-world problems.	L2 - Understand	
CO2	Experiment with data collection techniques, choosing fitting approaches to ensure sound research framework and methodology.	L3 - Apply	
CO3	Utilize research & analytic tools for enhancing the research publication	L2 - Understand	
CO4	Apply knowledge to produce presentations and technical reports that effectively communicate research findings.	L3 - Apply	
CO5	Explain types of intellectual property and comprehend patenting as essential for safeguarding innovation and creativity.	L2 - Understand	
	REFERENCE BOOKS		
1.	Cooper, Donald R.; Schindler, Pamela S.; Sharma, J. K., Business Research Methods, 12 th Edition, Tata McGraw Hill Education, 2012.		
2.	DePoy, Elizabeth; Gitlin, Laura N., Introduction to Research: Understanding and Applying Multiple Strategies, 6 th Edition, Elsevier Health Sciences, 2015.		
3.	Walliman, Nicholas, Research Methods: The Basics, 3 rd Edition, Routledge, 2017.		
4.	Bettig, Ronald V., Copyrighting Culture: The Political Economy of Intellectual Property, 1 st Edition, Routledge, 2018.		
5.	The Institute of Company Secretaries of India, Statutory body under an Act of parliament, "Professional Programme Intellectual Property Rights, Law and practice", September 2013.		
WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content
1.	Researchgate	https://www.researchgate.net/	Journal
2.	Degruyter	https://blog.degruyter.com/how-to-write-a-good-research-paper/	Web reference
3.	Elsevier	https://www.elsevier.com/en-in/researcher/author/submit-your-paper	Web reference



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VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	NPTEL	Edamana Prasad and Prathap Haridoss	Course Video	https://onlinecourses.nptel.ac.in/noc23_ge36/preview
2.	NPTEL	Feroz Ali	Course Video	https://onlinecourse.s.nptel.ac.in/noc22_hs59/preview

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2						2	3
CO2	2	3					2	2	3
CO3	2	3	2	3					3
CO4	2		3	3		3	2		3
CO5	2	2			3			3	3
Average	2.2	2.5	2.5	3.0	3.0	3	2	2.3	3
1-Low, 2-Medium, 3-High									


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MC23CA301	ADVANCED COMPUTER NETWORKS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A.- Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand the layered architecture and functionalities of computer networks.					
2.	To study application, transport, network, data link, and physical layer protocols in detail.					
3.	To analyze IPv4/IPv6 addressing, routing algorithms, and error/flow control techniques.					
4.	To explore wireless networks, mobile IP, and modern transmission techniques.					
5.	To introduce network management principles using SNMP.					
	INTRODUCTION (Not for Examination)					2
	Importance: The study of advanced computer networks is essential to understand the functioning of the Internet, communication protocols, and data transfer mechanisms. It enables students to design, analyze, and manage secure and efficient networks used in industries and research. Real-life Example(s): Everyday applications such as browsing websites using HTTP/HTTPS, sending emails, accessing cloud storage, video conferencing through Zoom or Teams, and using wireless networks like Wi-Fi or mobile data are all practical examples of advanced computer networks in action. Pre-requisite(s): Students are expected to have a basic understanding of computer fundamentals, operating systems, and data communication concepts. Familiarity with programming will also help in understanding protocol implementation and socket programming.					
UNIT-I	OVERVIEW OF THE INTERNET AND APPLICATION LAYER PROTOCOLS					9
	Overview of the Internet: Networks-Switching- The Internet-Accessing the Internet- Hardware and Software - Protocol Layering : TCP/IP Protocol Suite - The OSI Model - Application Layer: Introduction - Client-Server Paradigm - Application Programming Interface - Using Services of the Transport Layer Applications - *World Wide Web and HTTP- Electronic Mail-FTP - TELNET - Secure Shell (SSH) - Domain Name System (DNS) .					
UNIT-II	TRANSPORT LAYER PROTOCOLS AND TCP/IP SERVICES					9
	Transport-Layer: Introduction - Transport-Layer Protocols - Simple Protocol - Stop-and-Wait Protocol - Go-Back-N Protocol (GBN) - Selective-Repeat Protocol - Bidirectional Protocols: Piggybacking - Internet Transport-Layer Protocols - User Datagram Protocol - User Datagram - UDP Services - UDP Applications - Transmission Control Protocol - TCP Services -TCP Features - Segment - A TCP Connection - State Transition Diagram - Windows in TCP - Flow Control - Error Control -TCP Congestion Control - TCP Timers .					



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UNIT- III	NETWORK LAYER PROTOCOLS, IPV4/IPV6 AND ROUTING	9
	Introduction - Network Layer Protocols - IPv4 Datagram Format - IPv4 Addresses - Forwarding of IP Packets - ICMPv4 - Unicast Routing - General Idea - Routing Algorithms - Unicast Routing Protocols - Multicast Routing - Multicasting Basics - Intra-domain Routing Protocols - Inter-domain Routing Protocols - Next Generation IP - Packet Format - IPv6 Addressing - Transition from IPv4 to IPv6 - ICMPv6 .	
UNIT - IV	DATA LINK LAYER, WIRELESS NETWORKS AND MOBILE IP	9
	Data-Link Layer: Introduction - DLC - Framing - Flow and Error Control - Error Detection and Correction - Two DLC Protocols - MAC - Random Access - Controlled Access - Channelization - Link-Layer Addressing. Wireless Networks and Mobile IP: Wireless LANs - Introduction - IEEE 802.11 Project - Bluetooth - WiMAX - Other Wireless Networks - Channelization - Cellular Telephony - Satellite Networks - Mobile IP - Addressing - Agents - Three Phases - Inefficiency in Mobile IP .	
UNIT-V	PHYSICAL LAYER AND NETWORK MANAGEMENT	9
	Physical Layer: Data and Signals - Analog and Digital - Transmission Impairment - Data Rate Limits - Performance - Digital Transmission - Digital-to-Digital Conversion - Analog-to-Digital Conversion - Analog Transmission - Digital-to-Analog Conversion - Analog-to-Analog Conversion - Bandwidth Utilization - Multiplexing - Spread Spectrum - Transmission Media . Network Management: Introduction - SNMP - Managers and Agents - Management Components - An Overview - SMI - MIB - SNMP .	
Total		47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy
CO1	Explain the layered architecture of computer networks and the role of protocols.	L2 - Understand
CO2	Demonstrate the working of application and transport layer protocols including HTTP, FTP, DNS, TCP, and UDP.	L3 - Apply
CO3	Analyze network layer functionalities such as IPv4/IPv6 addressing, packet forwarding, and routing algorithms.	L4 - Analyze
CO4	Apply data link layer concepts, error detection/correction, and evaluate wireless/mobile networking techniques.	L3 - Apply
CO5	Illustrate physical layer transmission methods and network management using SNMP.	L3 - Apply
TEXTBOOKS		
1.	Behrouz A Forouzan, Firouz Mosharraf, "Computer Networks: A top down Approach", McGraw Hill Education, Special Indian Edition 2012.	
2.	James F Kurose and Keith W Ross, "Computer Networking: A Top - Down Approach", Pearson Education; 7th Edition (2017).	



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REFERENCE BOOKS	
1.	Kevin R. Fall, W. Richard Stevens, "TCP/IP Illustrated, Volume 1-The Protocols", Pearson Education, 2 nd Edition (2014).
2.	Larry Peterson, Bruce Davie, "Computer Networks, A systems Approach", Morgan Kaufmann Publishers, 5 th Edition (2011).
3.	Uyless Black, "Computer Networks: Protocols, Standards and Interface", Prentice Hall India Learning Private Limited, 8 th Edition (2015).
4.	William Stallings, "Foundations of Modern Networking: SDN, NFV, QoE, IoT, and Cloud", Pearson Education, 1 st Edition (2016).
5.	The Illustrated Network: How TCP/IP Works in a Modern Network 2nd edition Walter Goralski Morgan Kaufmann Publications.

WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content
1.	Geeksforgeeks	https://www.geeksforgeeks.org/computer-networks/computer-network-tutorials/	Web References
2.	Tutorialspoint	https://www.tutorialspoint.com/computer_fundamentals/computer_networking.htm	Web References
3.	HackerRank	https://www.hackerrank.com/skills-directory/networking	Web References

VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Crash Course	Lecture	https://www.youtube.com/watch?v=3QhU9jd03a0
2.	NPTEL	Prof. Sujoy Ghosh	Lecture	https://archive.nptel.ac.in/course.html
3.	YouTube	Kunal Kushwaha	Lecture	https://www.youtube.com/watch?v=IPvYjXCstg8

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2		2					
CO2	3	3	2	3					
CO3	3	3	3	3					
CO4	3	3	3	3					
CO5	3	2		2					
Average	3	2.6	2.7	2.6					
1-Low, 2-Medium, 3-High									

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MC23CA302	DATA STRUCTURES AND ALGORITHMS USING C	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A. - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand the types of data structures and implement list ADT.					
2.	To design algorithm using linear data structures like stack and queue.					
3.	To solve the problems using non-linear data structure tree.					
4.	To elucidate the problems using non-linear data structure graph.					
5.	To compare efficiency of various sorting and searching techniques.					
	INTRODUCTION (Not for Examination)					2
	Importance: Efficiency in Data Management – Performance Optimization – Real-World Applications – Competitive Programming and Contest – Problem Solving Skills. Real-life Example(s): Arrays: Online Shopping Carts – Linked Lists: Music Playlists – Stacks: Web Browser History – Queues: Customer Service Systems – Trees: File Systems – Graphs: Social Networks and Google Maps. Pre-requisite: Problem Solving using C Programming – Computer Organization – Design Thinking.					
UNIT-I	INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS					9
	Introduction – Need for Data Structures – Classification – Abstract Data Types – Algorithm Analysis : Time Complexity and Space Complexity – Asymptotic Notation (Big O, Omega, Theta) .					
UNIT-II	LINEAR DATA STRUCTURES					9
	Arrays – Operations (Insertion, Deletion, Searching, Sorting) – Linked List : Types (Singly, Doubly, Circular) – Stack: Operations and Applications (Infix to Postfix Conversion, Expression Evaluation) – Queue: Types (Simple Queue, Circular Queue, Deque, Priority Queue) – Applications .					
UNIT- III	NON-LINEAR DATA STRUCTURE: TREES					9
	Trees: Definition – Basic Terminologies – Types (Binary Tree, Binary Search Tree, AVL Tree) – Operations (Insertion, Deletion, Traversal, Searching) .					
UNIT - IV	NON-LINEAR DATA STRUCTURE: GRAPHS					9
	Graphs: Definition – Basic Terminologies – Representation – Graph Traversal Algorithms (BFS and DFS) – Graph Algorithms: Topological Sort – Dijkstra’s Algorithm – Minimum Spanning Tree Algorithms (Kruskal’s and Prim’s Algorithm) .					
UNIT-V	SORTING, SEARCHING ALGORITHMS AND HASHING					9
	Sorting Algorithms: Bubble Sort , Selection Sort , Insertion Sort , Merge Sort , Quick Sort , Heap Sort – Searching Algorithms: Linear Search , Binary Search – Hashing: Hash Tables - Hashing Techniques .					

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		Total	47 Periods	
	OPEN-ENDED PROBLEMS / QUESTIONS			
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Course Outcomes: Upon completion of this course, the students will be able to:			BLOOM'S Taxonomy	
CO1	Understand the fundamentals of data structures, their classification, and the need for efficient data organization.		L2- Understand	
CO2	Analyze and evaluate algorithm performance using time and space complexity, including asymptotic notations.		L4 - Analyze	
CO3	Implement linear and non-linear data structures such as arrays, linked lists, stacks, queues, trees, and graphs.		L3- Apply	
CO4	Apply sorting, searching, and hashing techniques for optimized data processing and retrieval.		L3- Apply	
CO5	Develop efficient algorithmic solutions for real-world computational problems using appropriate data structures.		L6- Create	
TEXTBOOKS				
1.	Reema Thareja, "Data Structures Using C", 2 nd Edition, Oxford University Press, 2014.			
2.	Yashavant Kanetkar, "Data Structures Through C", BPB Publications, 2020.			
REFERENCE BOOKS				
1.	Mark Allen Weiss, "Data Structures and Algorithm Analysis in C", 2 nd Edition, Pearson Education, 2002.			
2.	Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, "Fundamentals of Data Structures in C", 2 nd Edition, Universities Press, 2008.			
3.	Narasimha Karumanchi, "Data Structures and Algorithms Made Easy", 5 th Edition, CareerMonk Publications, 2016.			
WEB REFERENCES				
S. No.	Publisher	Website link	Type of Content	
1.	TutorialsPoint	https://www.tutorialspoint.com/dsa_using_c/dsa_using_c_useful_resources.htm	Web Reference	
2.	HackerRank	https://www.hackerrank.com/domains/data-structures	Problem Solving	
3.	GeeksforGeeks	https://www.geeksforgeeks.org/real-time-application-of-data-structures/	Web Reference	
VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	K. Ravikumar	Lecture	https://www.youtube.com/@reachtutorravi3115
2.	YouTube	Jenny's Lectures	Lecture	https://www.youtube.com/watch?v=AT14lCXuMKI&list=PLdo5W4Nhv31bbKJzrsKfMpo_grxuLI8LU&pp=iAQB

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3.	NPTEL	Sudarshan Iyengar	Lecture	https://onlinecourses.archive.nptel.ac.in/noc18_cs25/preview
4.	YouTube	K. Ravikumar	Lecture	https://www.youtube.com/@reachtutorravi3115

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2							
CO2	3	3							
CO3	3	3	3	2					
CO4	3	3	3	3					
CO5	3	3	3	3					
Average	3	2.8	3	2.7					
1-Low, 2-Medium, 3-High									



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MC23CA303	ADVANCED DATABASE TECHNOLOGY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A. - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand relational database systems, including their architecture, design, and implementation.					
2.	To grasp relational database design principles, with a focus on functional dependencies and normalization techniques.					
3.	To provide students with practical knowledge and hands-on experience in SQL and PL/SQL for effective database management and querying.					
4.	To provide students with a thorough understanding of transaction management and database security in relational database systems.					
5.	To emphasize normalization, query optimization, database security, and modern database trends like MongoDB.					
	INTRODUCTION (Not for Examination)					2
	Importance: Way to organize and manage large volumes of data efficiently, prevent unauthorized access and ensure data consistency, enable fast and efficient retrieval of data through querying languages like SQL, manage concurrent access to the data by multiple users or applications, Ensure the data available and protect against loss or corruption, supports decision-making process by providing tools for data analysis, reporting and visualization. Real-life Example(s): University Database – Library Management System – Inventory Management for EMart Grocery Shop – Banking System. Linkages: Pre-requisite: Problem Solving using C++.					
UNIT-I	RELATIONAL DATABASE					9
	Introduction - Purpose of Database System — Views of data — Database Architecture — Structure of Relational Databases - Database Schema - Keys - Schema Diagrams - Entity-Relationship (ER) Model - Mapping Cardinalities and Participation Constraints - Reduction to Relational Schemas - Entity-Relationship Design Issues - Extended ER Features .					
UNIT-II	DATABASE DESIGN					9
	Features of Good Relational Designs - Functional Dependencies — Non-loss Decomposition — First, Second, Third Normal Forms — Boyce/Codd Normal Form — Multi-valued Dependencies and Fourth Normal Form — Join Dependencies and Fifth Normal Form - Sixth Normal Form - Denormalization in databases - Refactoring Databases for Performance .					
UNIT- III	STRUCTURED QUERY LANGUAGE					9
	SQL Data Definition and Data Types - SQL Constraints - Data Definition Language (DDL) Commands - Data Manipulation Language (DML) Commands - Data Query Language (DQL) commands - Database Querying – Aggregate Functions - Set Operations - Nested					

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	Sub queries – Joins – Views and Indexes – Query Execution Plan Analysis - PL/SQL - Functions and Procedures – Triggers.	
UNIT – IV	TRANSACTION MANAGEMENT	9
	Transaction Concepts – ACID Properties – Schedules – Serializability – Concurrency Control – Need for Concurrency – Locking Protocols – Two Phase Locking – Deadlock – Transaction Recovery – Save Points – Transaction Control Language (TCL) commands in SQL - Isolation Levels – Database security – Access control – Data Control Language (DCL) commands in SQL - Backup and recovery .	
UNIT-V	NoSQL and MongoDB	9
	Relational Vs Non-relational database - Introduction to NoSQL – Types of NoSQL Databases - CAP Theorem – Sharding . MongoDB: Introduction – Database , Collections and Documents in MongoDB - JavaScript Object Notation - Datatypes - Data Modelling - Basic Operations: Insert, Update, Delete , Query – Aggregation , Indexing and Joining collections - Backup and recovery .	
Total		47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy
CO1	Understand the fundamental concepts of relational databases, database architecture, and the Entity-Relationship model.	L2- Understand
CO2	Design and normalize relational databases to ensure data integrity and efficiency.	L4 – Analysis
CO3	Develop and execute queries using SQL, including DDL, DML, DQL, and advanced SQL features like joins, views, functions, and triggers.	L6 – Create
CO4	Implement transaction management techniques, concurrency control, and database recovery mechanisms.	L3 – Apply
CO5	Explore NoSQL databases, particularly MongoDB, and perform data modelling, aggregation, indexing, and backup operations.	L3 – Apply
TEXTBOOKS		
1.	R. Elmasri, S.B. Navathe, "Fundamentals of Database Systems", 7 th Edition, Pearson Education/Addison Wesley, 2017.	
2.	Abraham Silberschatz, Henry F Korth, S. Sudharshan, "Database System Concepts", 7 th Edition, McGraw Hill, 2019.	
REFERENCE BOOKS		
1.	C. J. Date, A. Kannan, S. Swamynathan, "An Introduction to Database Systems", 8 th Edition, Pearson Education, 2006.	
2.	Brad Dayley, "Teach Yourself NoSQL with MongoDB in 24 Hours", Sams Publishing, 1 st Edition, 2014.	
3.	Manu Sharma, "MongoDB Complete Guide", BPB Publications, 2021.	

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WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content
1.	MYSQL	https://dev.mysql.com/doc/	Web References
2.	W3Schools	https://www.w3schools.in/dbms	Web References
3.	IGI Global	https://www.igiglobal.com/journal/journal-database-management/1072	Web References

VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Sundeeep Saradhi Kanthety	Lecture	https://www.youtube.com/watch?v=WFX5azcMHFM
2.	YouTube	Abhishek	Lecture	https://www.youtube.com/watch?v=xoTyrdT9SZI&list=PLLGlmW7JT-nTr1ory9o2MgsOmmx2w8FB3
3.	YouTube	Net Ninja	Lecture	https://www.youtube.com/watch?v=ExcRbA7fy_A&list=PL4cUxeGkcC9h77dJ-QJlwGlZITd4ecZOA

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2		1					
CO2	3	3	2						
CO3	3	2	3	2					
CO4	2	3	3	2					
CO5	2	2	3	3					
Average	2.6	2.4	2.75	2					
1-Low, 2-Medium, 3-High									

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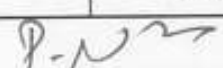
MC23CA304	OBJECT-ORIENTED PROGRAMMING USING JAVA	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand the fundamentals of Java, including its syntax, data types, control structures, and the Java Virtual Machine.					
2.	To apply core object-oriented concepts such as encapsulation, inheritance, polymorphism, and abstraction in Java.					
3.	To develop Java applications with features like exception handling, multithreading, generics, and collections.					
4.	To perform file operations, serialization, and leverage modern Java features like lambda expressions and the Stream API.					
5.	To build graphical user interface (GUI) applications using Java Swing and implement database connectivity using JDBC.					
	INTRODUCTION (Not for Examination)					2
	Importance: Platform Independence - Object-Oriented Programming - Enterprise Application - Android Application Development. Real-life Example(s): Online banking system - Mobile applications - E-commerce websites. Pre-requisite: • Basic knowledge of programming concepts • Understanding of fundamental programming logic • Familiarity with C or any other procedural programming language					
UNIT-I	INTRODUCTION TO JAVA PROGRAMMING					9
	Introduction to Object-Oriented Programming (OOP) - OOP Concepts Realization - Overview of Java - Features of Java - Java Virtual Machine (JVM) - JDK , JRE - Environmental Setup - Compiling and executing using Command Line and IDE - Basic Syntax - Data Types - Operators - Control Flow Statements - Arrays and Strings - Functions.					
UNIT-II	OOP CONCEPTS IN JAVA					9
	Classes and Objects - Access Modifiers - Constructors - Method Overloading - Static - Inner Classes (member, local, anonymous) - Encapsulation - Inheritance - Types of Inheritance - Method Overriding - Super and This Keyword - Abstract Classes and Interfaces - Packages .					
UNIT- III	EXCEPTION HANDLING, MULTITHREADING AND COLLECTIONS FRAMEWORK					9
	Exception Handling: Try, Catch, Finally, Throw, Throws - Built-in and User -defined Exceptions - Multithreading: Thread Lifecycle - Creating and Running Threads - Synchronization - Inter-Thread Communication - Generics in Java - Collections Framework: ArrayList, LinkedList, HashSet, TreeSet, HashMap, TreeMap .					

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UNIT – IV	FILE HANDLING AND JAVA NEW FEATURES	9	
	File Handling: Reading and Writing Files – Streams (Byte and Character) – Serialization – Java New Features: Lambda Expressions – Stream API – New Date and Time API .		
UNIT-V	GUI PROGRAMMING AND DATABASE CONNECTIVITY	9	
	Event-Driven Programming – Java Swing Components (JFrame, JButton, JTextField, JLabel, etc.) - Event Handling: Listeners and Adapters – Database Connectivity (JDBC): Connecting Java with MySQL – Executing Queries – PreparedStatement .		
Total		47 Periods	
OPEN-ENDED PROBLEMS / QUESTIONS			
Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.			
Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy	
CO1	Understand the fundamental concepts of Object-Oriented Programming and how they are implemented in Java.	L2- Understand	
CO2	Develop Java programs using classes, objects, inheritance, polymorphism, and encapsulation.	L3 – Apply	
CO3	Apply exception handling mechanisms and multi-threading concepts.	L3 – Apply	
CO4	Implement file handling, collections framework, and database connectivity using JDBC.	L3 – Apply	
CO5	Design and develop GUI-based applications using Java Swing and event handling.	L6 – Create	
TEXTBOOKS			
1.	Herbert Schildt, Java: The Complete Reference, 11th Edition, McGraw Hill, 2018.		
2.	Cay S. Horstmann, Core Java Volume I – Fundamentals, 11th Edition, Pearson, 2018.		
REFERENCE BOOKS			
1.	E. Balagurusamy, Programming with Java, 6th Edition, McGraw Hill, 2019.		
2.	Y. Daniel Liang, Introduction to Java Programming and Data Structures, 11th Edition, Pearson, 2017.		
3.	Kathy Sierra, Bert Bates, Head First Java, 2nd Edition, O'Reilly, 2005.		
WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content
1.	Web Reference	https://webreference.com/ java/	Web Reference
2.	w3schools	https://www.w3schools.com /java/	Web Reference

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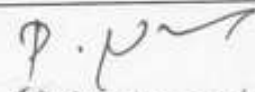

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3.	Web based programming	http://www.webbased programming.com/Java-Developers-Reference/		Web Reference
VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	CodeWithHarry	Lecture	https://www.youtube.com/watch?v=8cm1x4bC610
2.	YouTube	Mosh Hamedani	Lecture	https://www.youtube.com/watch?v=eIrMbaQSU34
3.	YouTube	freeCodeCamp.org	Lecture	https://www.youtube.com/watch?v=grEKMHGyyns

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2	1						
CO2	3		3	2					
CO3			2	2					
CO4			2	3					
CO5			2	3					
Average	3	2	2	2.5					
1-Low, 2-Medium, 3-High									


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MC23CA305	DATA STRUCTURES AND ALGORITHMS LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.C.A. - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To acquire the knowledge of using advanced tree structures					
2.	To learn the usage of heap structures					
3.	To understand the usage of graph structures and spanning trees					
4.	To identify the problems such as matrix chain multiplication, activity selection and Huffman coding					
5.	To understand the necessary mathematical abstraction to solve problems.					
LIST OF EXPERIMENTS						
1. Implementation of recursive function for tree traversal and Fibonacci 2. Implementation of iteration function for tree traversal and Fibonacci 3. Implementation of Merge Sort and Quick Sort 4. Implementation of a Binary Search Tree 5. Red-Black Tree Implementation 6. Heap Implementation 7. Fibonacci Heap Implementation 8. Graph Traversals 9. Spanning Tree Implementation 10. Shortest Path Algorithms (Dijkstra's algorithm, Bellman Ford Algorithm) 11. Implementation of Matrix Chain Multiplication 12. Activity Selection and Huffman Coding Implementation						
HARDWARE/SOFTWARE REQUIREMENTS						
1. 64-bit Open-Source Linux or its derivative 2. Open-Source C++ Programming tool like G++/GCC						
Total (P)						60 Periods
Course Outcomes: Upon completion of this course, the students will be able to:						BLOOM'S Taxonomy
CO1	Design and implement basic and advanced data structures extensively					L3 – Apply
CO2	Plan algorithms using graph structures					L3 – Apply
CO3	Approach and develop efficient algorithms with minimum complexity using design techniques					L3 – Apply
CO4	Develop programs using various algorithms					L3 – Apply
CO5	Choose appropriate data structures and algorithms, understand the ADT/libraries, and use it to design algorithms for a specific problem.					L4 – Analyze


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REFERENCE BOOKS	
1.	Lipschutz Seymour, "Data Structures Schaum's Outlines Series", Tata McGraw Hill, 3rd Edition, 2014.
2.	Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, "Data Structures and Algorithms", Pearson Education, Reprint 2006.
3.	http://www.coursera.org/specializations/data-structures-algorithms
4.	http://www.tutorialspoint.com/data_structures_algorithms

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3		3	2					
CO2		3	2						
CO3	2	3	3						
CO4			3	2					
CO5		2	3	2					
Average	2.5	2.7	2.8	2					
1-Low, 2-Medium, 3-High									

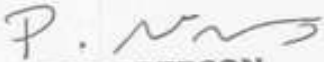


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MC23CA306	ADVANCED DATABASE TECHNOLOGY LABORATORY	CP 4	L 0	T 0	P 4	C 2
Programme & Branch	M.C.A. - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand the process of distributing tables across multiple systems					
2.	To know the process of storing, retrieving spatial and temporal data					
3.	To escalate the process of storing, retrieving objects in a database					
4.	To grasp the process of storing and retrieving data from a XML Database					
5.	To use the open-source database for building a mobile application					
LIST OF EXPERIMENTS						
1. NOSQL Exercises a. MongoDB – CRUD operations, Indexing, Sharding b. Cassandra: Table Operations, CRUD Operations, CQL Types c. HIVE: Data types, Database Operations, Partitioning – HiveQL d. OrientDB Graph database – OrientDB Features 2. MySQL Database Creation, Table Creation, Query 3. MySQL Replication – Distributed Databases 4. Spatial data storage and retrieval in MySQL 5. Temporal data storage and retrieval in MySQL 6. Object storage and retrieval in MySQL 7. XML Databases, XML table creation, XQuery FLWOR expression 8. Mobile Database Query Processing using open source DB (MongoDB/MySQL etc)						
HARDWARE/SOFTWARE REQUIREMENTS						
1. Java / Python / R / Scala 2. Oracle, MySQL, MongoDB, Casandra, Hive						
					Total (P)	60 Periods
	Course Outcomes: Upon completion of this course, the students will be able to:					BLOOM'S Taxonomy
CO1	Design and implement advanced databases					L3 – Apply
CO2	Use big data frameworks and tools.					L3 – Apply
CO3	Formulate complex queries using SQL.					L3 – Apply
CO4	Create an XML document and perform Xquery					L3 – Apply
CO5	Query processing in Mobile databases using open source tools.					L4 – Analyze

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Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3		3	2					
CO2		3	2	2					
CO3	3	2	3						
CO4			2	2					
CO5		2	2	2					
Average	3	2.3	2.6	2					
1-Low, 2-Medium, 3-High									

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MC23CA307	OBJECT-ORIENTED PROGRAMMING USING JAVA LABORATORY	CP 4	L 0	T 0	P 4	C 2
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To build software development skills using java programming for real-world applications.					
2.	To understand and apply the concepts of classes, packages, interfaces, inheritance, exception handling and file processing.					
3.	To develop applications using generic programming and event handling					
LIST OF EXPERIMENTS						
1. Solve problems by using sequential search, binary search, and quadratic sorting algorithms (selection,insertion) 2. Develop stack and queue data structures using classes and objects. 3. Develop a java application with an Employee class with Emp_name, Emp_id, Address, Mail_id, Mobile_no as members. Inherit the classes, Programmer, Assistant Professor, Associate Professor and Professor from employee class. Add Basic Pay (BP) as the member of all the inherited classes with 97% of BP as DA, 10 % of BP as HRA, 12% of BP as PF, 0.1% of BP for staff club funds. Generate pay slips for the employees with their gross and net salary. 4. Write a Java Program to create an abstract class named Shape that contains two integers and an empty method named printArea (). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method printArea () that prints the area of the given shape . 5. Solve the above problem using an interface. 6. Implement exception handling and creation of user defined exceptions. 7. Write a java program that implements a multi-threaded application that has three threads. First thread generates a random integer every 1 second and if the value is even, the second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of the cube of the number. 8. Write a program to perform file operations. 9. Develop applications to demonstrate the features of generics classes. 10. Develop applications using JavaFX controls, layouts and menus. 11. Develop a mini project for any application using Java concepts .						
HARDWARE/SOFTWARE REQUIREMENTS						
1. Lab Requirements: for a batch of 30 students 2. Operating Systems: Linux / Windows 3. Front End Tools: Eclipse IDE / Netbeans IDE.						
Total (P)						60 Periods
Course Outcomes: Upon completion of this course, the students will be able to:						BLOOM'S Taxonomy
CO1	Design and develop java programs using object oriented programming concepts					L3 – Apply
CO2	Develop simple applications using object oriented concepts such as package, exceptions					L3 – Apply
CO3	Implement multithreading, and generics concepts					L3 – Apply

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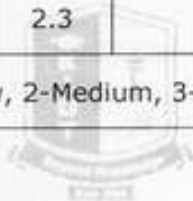
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CO4	Create GUIs and event driven programming applications for real world problems	L4 – Analyze
CO5	Implement and deploy web applications using Java	L3– Apply

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3		3						
CO2			2	2					
CO3			3	2					1
CO4			2			2			
CO5			3	3			2		
Average	3		2.6	2.3		2	2		1
1-Low, 2-Medium, 3-High									



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Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy
CO1	Understand and apply basic directory and file management commands in Linux.	L2 – Understand
CO2	Use general-purpose, filter, and pattern-matching commands for effective data handling.	L3 – Apply
CO3	Perform process management and system monitoring using Linux utilities.	L2 – Understand
CO4	Develop shell scripts for automating routine tasks and problem-solving.	L3 – Apply
CO5	Build practical skills for real-time Linux system operations in academic and industrial contexts.	L4 – Analyze

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2	2	1					
CO2	3	3	2	2					
CO3	3	3	3	2					
CO4	2	3	3	3					
CO5	2	2	2	3					
Average	2.6	2.6	2.4	2.2					
1-Low, 2-Medium, 3-High									

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MC23PT701	COMMUNICATION SKILLS ENHANCEMENT - I	CP	L	T	P	C
		2	0	0	2	NC
Programme & Branch	MASTER OF COMPUTER APPLICATIONS (M.C.A)	Version: 1.0				
Course Objectives:						
1.	To develop listening skills through real-life communication contexts.					
2.	To construct spoken communication content for academic, professional, and social contexts.					
3.	To apply reading techniques to enhance comprehension and critical analysis.					
4.	To build academic and professional writing skills to ensure clarity, structure, and purpose.					
	INTRODUCTION (Not for Examination)					2
Importance	LSRW skills are focused in improving comprehension, fluency, vocabulary and writing for academic, professional and every day communication. These skills prepare students for career opportunities and effective interaction in various contexts.					
Real-life Example(s)	(i) Software engineers use communications skills to interact with clients (ii) English language skills are very much essential for Writing project proposals and documents at a work place					
Linkages	Communication Skills Enhancement - II					
	LIST OF ACTIVITIES					
	LISTENING					7
1.	Listening and practicing neutral accents					
2.	Understanding short talks and lectures with listening exercises					
3.	Listening to TED Talks/ dialogues					
	SPEAKING					8
4.	Role plays and extempore speaking					
5.	Participating in Group Discussions					
6.	Making Presentations					
7.	Expressing oneself in day to day situation					
	READING					7
8.	Understanding the techniques of Reading to develop comprehension and analysis, Skimming and Scanning					
9.	Reading Brochure, Company Profile, etc.					
10.	Reading Newspapers and articles.					

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M.E. / M.Tech / M.C.A Regulations 2023

	WRITING		8	
11.	Verbal Analogies			
12.	Spotting Errors			
13.	Formal and Informal Letter Writing / Email Writing			
	Total (P)		32 Periods	
	Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy	
CO1	Apply listening skills to understand conversation and respond to it.		L3 - Apply	
CO2	Make presentations and participate in discussions.		L3 - Apply	
CO3	Apply reading strategies to comprehend academic and professional texts.		L3 - Apply	
CO4	Use vocabulary and grammar to construct error-free writing.		L3 - Apply	
S.No.	TEXTBOOKS:			
1.	Murphy, Raymond. English Grammar in use (5 th Edition). Cambridge Press: New York, 2019.			
2.	Richardson, Mathew. Advanced Communication Skills. Charlie Creative Lab, 2020.			
S.No.	REFERENCE BOOKS			
1.	Comfort, Jeremy, et al. Speaking Effectively: Developing Speaking Skills for Business English. Cambridge University Press, Cambridge: Reprint 2011.			
2.	Azar, B.S. Understanding and Using English Grammar (4 th edition). Pearson Education, 2009.			
3.	Prabhavathi, Y. People Interface: English Language Communication Skills – Manual/Workbook. Cengage Learning India, 2018.			
	WEB REFERENCES			
S.No.	Publisher	Website link	Type of Content	
1.	Englisch-hilfen.de	https://www.englisch-hilfen.de/en/	Grammar & Vocabulary	
2.	FluentU	https://www.fluentu.com/	Language Learning through videos	
	VIDEO REFERENCES			
S.No.	Video Details	Name of the Expert	Type of Content	Video link
1.	SWAYAM	Dr. Shoba. K.N.	Communication Skills in English	https://onlinecourses.swayam2.ac.in/ntr25_ed101/preview
2.	YouTube	Simerjeet Singh	Master the Art of Communication Skills	https://www.youtube.com/watch?v=mloCf86Q_ow&list=PLOaeOd121eBEEWP14TYgSnFsvaTI PD22

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S.No.	ONLINE COURSES	
1.	https://www.open.edu/openlearn/money-business/effective-communication-the-workplace?active-tab=description-tab	
2.	https://www.life-global.org/categorylist/1-communication	

Mapping of COs with POs and PSOs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1						2		1	
CO2						2		1	
CO3						2		1	
CO4						2		1	
Avg.						2.0		1.0	
1-Low, 2 -Medium, 3-High.									


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SEMESTER – II




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MC23CA309	CYBERSECURITY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To learn the principles of cybersecurity and to identify threats and risks.					
2.	To learn how to secure physical assets and develop system security controls.					
3.	To understand how to apply security for Business applications and Network Communications.					
4.	To learn the technical means to achieve security.					
5.	To learn to monitor and audit security measures.					
	INTRODUCTION (Not for Examination)					2
Importance	Cybersecurity is crucial in today's digital age because it protects sensitive information, financial transactions, and critical infrastructure from cyber threats. It's vital for maintaining business continuity, safeguarding personal data, and ensuring the overall security of individuals and organizations in an increasingly interconnected world.					
Real-life Example(s)	Anti-virus software – Firewalls - Encryption					
Linkages	Pre-requisite: Networking, and security Future Courses: Blockchain Technology.					
UNIT-I	PLANNING FOR CYBERSECURITY					9
	Best practices including standards and plan of action, security governance principles, components and approach, information risk management, asset identification, threat identification, vulnerability identification, risk assessment approaches, likelihood and impact assessment, risk determination, evaluation and treatment, security management function, security policy and acceptable use policy, security management best practices, and security models such as Bell-LaPadula model, Biba Integrity Model, and Chinese Wall model.					
UNIT-II	SECURITY CONTROLS					9
	People management, human resource security, security awareness and education, information management, information classification and handling, privacy, documents and record management, physical asset management, office equipment security, industrial control systems security, mobile device security, system development, incorporating security into the SDLC, and disaster management and incident response planning.					
UNIT- III	CYBERSECURITY FOR BUSINESS APPLICATIONS AND NETWORKS					9
	Business application management, corporate business application security, end-user developed applications, system access, authentication mechanisms, access control, system management,					

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	virtual servers, network storage systems, network management concepts, firewall, IP security, electronic communications, and a case study on OWASP vulnerabilities using the OWASP ZAP tool.	
UNIT – IV	TECHNICAL SECURITY	9
	Supply chain management, cloud security, security architecture, malware protection, intrusion detection, digital rights management, cryptographic techniques, threat and incident management, vulnerability management, security event management, forensic investigations, local environment management, and business continuity.	
UNIT-V	SECURITY ASSESSMENT	9
	Security monitoring and improvement, security audit, security performance, information risk reporting, information security compliance monitoring, and security monitoring and improvement best practices.	
	Total(L)	47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Develop a set of risk and security requirements to ensure that there are no gaps in an organization's security practices.	L3 - Apply
CO2	Achieve management, operational and technical means for effective cybersecurity.	L3 - Apply
CO3	Audit and monitor the performance of cybersecurity controls.	L3 - Apply
CO4	Spot gaps in the system and devise improvements.	L3 - Apply
CO5	Identify and report vulnerabilities in the system	L3 - Apply
	TEXT BOOKS	
1.	William Stallings, "Effective Cybersecurity - A guide to using Best Practices and Standards", Addison-Wesley Professional, First Edition, 2019.	
2.	Raef Meeuwisse, "Cybersecurity for Beginners", Second Edition, Cyber Simplicity Ltd, 2017.	


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REFERENCE BOOKS	
1.	Patrick Engebretson, "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made Easy", 2nd Edition, Syngress, 2013.
2.	Adam Shostack, "Threat Modelling - Designing for Security", Wiley Publications, First Edition, 2014.
3.	Gregory J. Touhill and C. Joseph Touhill, "Cybersecurity for Executives - A Practical Guide", Wiley Publications, First Edition, 2014.
4.	Charles P. Pfleeger, Shari Lawrence Pfleeger, Jonathan Margulies, "Security in Computing", Fifth Edition, Prentice Hall, 2015.

WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content
1.	NIST Cybersecurity Framework & Standards	https://www.nist.gov/cyberframework	Web References
2.	Cisco Networking Academy – Free Cybersecurity Modules	https://www.netacad.com	Web References
3.	Microsoft Learn – Free Security Training	https://learn.microsoft.com/en-us/training/browse/?products=security	Web References

VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Cybersecurity Full Course – 8 Hours	Lecture	https://www.youtube.com/watch?v=inWWhr5tnEA
2.	NPTEL	NPTEL – Introduction to Cybersecurity	Lecture	https://nptel.ac.in/courses/106/104/106104223
3.	YouTube	Security Auditing & Monitoring	Lecture	https://www.youtube.com/watch?v=Bqg8Bb6zq2c

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2		1					1
CO2	3	3	2		1			1	
CO3	2	3	3	2			1		
CO4		2	3	3	2			1	
CO5	3		3	2	3				1
Average	2.8	2.5	2.8	2.0	2.0		1.0	1.0	1.0

1-Low, 2-Medium, 3-High

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UX23MC701	UNIVERSAL HUMAN VALUES AND ETHICS	CP	L	T	P	C
		3	1	0	2	NC
Programme & Branch	COMMON TO M.E. (SE,AE,PED,ISE), M.C.A & MBA (IEV)	Version: 1.1				
Course Objectives:						
1.	To understand the concept of Universal Human Values.					
2.	To discuss theoretical and practical implications of UHV.					
3.	To relate the use of harmony in the family and society.					
4.	To classify the harmony in the nature methods.					
5.	To construct effective human values in personal and professional in life.					
UNIT-I	INTRODUCTION TO VALUE EDUCATION					3+6
	Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Understanding Value Education - Sharing about Oneself - Self-exploration as the Process for Value Education - Continuous Happiness and Prosperity - the Basic Human Aspirations - Exploring Human Consciousness - Happiness and Prosperity - Current Scenario - Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance . * Experiential Learning: Practice sessions to discuss natural acceptance in human being.					
UNIT-II	HARMONY IN THE HUMAN BEING					3+6
	Understanding Human being as the Co-existence of the Self and the Body - Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self and Body - The Body as an Instrument of the Self - Understanding Harmony in the Self - Exploring Sources of Imagination in the Self- Harmony of the Self with the Body - Programme to ensure self-regulation and Health - Exploring Harmony of Self with the Body . * Experiential Learning: Practice sessions to discuss the awareness of self, help in managing stress and anxiety.					
UNIT-III	HARMONY IN THE FAMILY AND SOCIETY					3+6
	Harmony in the Family - the Basic Unit of Human Interaction - 'Trust' - the Foundational Value in Relationship - Exploring the Feeling of Trust - 'Respect' - as the Right Evaluation - Exploring the Feeling of Respect - Other Feelings , Justice in Human-to-Human Relationship - Understanding Harmony in the Society - Vision for the Universal Human Order - Exploring Systems to fulfil Human Goal. * Experiential Learning: Case Studies on Family and societal issues.					

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UNIT – IV	HARMONY IN THE NATURE/EXISTENCE	3+6
	Understanding Harmony in the Nature – Interconnectedness , self-regulation and Mutual Fulfilment among the Four Orders of Nature - Exploring the Four Orders of Nature - Realizing Existence as Co-existence at All Levels - The Holistic Perception of Harmony in Existence - Exploring Co-existence in Existence. * Experiential Learning: Case Study to discuss human being as cause of imbalance in nature, pollution, depletion of resources and role of technology.	
UNIT – V	IMPLICATIONS OF THE HOLISTIC UNDERSTANDING - A LOOK AT PROFESSIONAL ETHICS	3+6
	Natural Acceptance of Human Values - Definitiveness of (Ethical) Human Conduct - Exploring Ethical Human Conduct - A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order - Competence in Professional Ethics - Exploring Humanistic Models in Education - Holistic Technologies, Production Systems and Management Models -Typical Case Studies - Strategies for Transition towards Value-based Life and Profession - Exploring Steps of Transition towards Universal Human Order. * Experiential Learning: Case studies of holistic technologies and Professional ethics.	
Total (LP) : 45 Periods		
* Internal Evaluation only.		
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the class room teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Interpret the concepts of Universal Human Values.	L2 - Understand
CO2	Summarize both theoretical and practical implications of Universal Human Values.	L2 - Understand
CO3	Build the harmony in family and society.	L3 - Apply
CO4	Practice harmony in all human existence.	L3 - Apply
CO5	Relate human values in both personal and professional life.	L2 - Understand
	REFERENCE BOOKS:	
1.	R.R Gaur, R Sangal, G P Bagaria, A foundation course in Human Values and professional Ethics – Teachers Manual, Excel books, New Delhi	

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2.	B L Bajpai, 2004, Indian Ethos and Modern Management, New Royal Book Co., Lucknow, Reprinted 2008.		
3.	Frankl, Viktor E. Yes to Life In spite of Everything, Penguin Random House, London, 2019.		
4.	Van Zomeren, M., & Dovidio, J. F. The Oxford Handbook of the Human Essence (Eds.), New York Oxford University Press, 2018.		
5.	R R Gaur, R Asthana, G P Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019.		
6.	A.N. Tripathi, Human Values, New Age Intl. Publishers, New Delhi, 2004.		
S.No.	WEB REFERENCES:		
1.	https://www.researchgate.net/publication/382495858_Universal_Human_Value		
2.	https://fdp-sl.aicte-india.org/UHVII.php		
	VIDEO REFERENCES:		
S.No.	Publisher	Website link	Type of Content
1.	Dr.Rajneesh Arora	https://www.youtube.com/watch?v=SdXIB0k7klQ&list=PLWDeKF97v9SP7wSlapZcQRrT7OH0ZIGC4&index=2	Video Lecture
2.	Mr.Umesh Jadhav	https://www.youtube.com/watch?v=_g8Hw2APx8Q	Video Lecture

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Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1						1	1	3	2
CO2						2	1	3	2
CO3						2	3	3	2
CO4						1	2	3	2
CO5						2	2	3	3
Average						1.6	1.8	3	2.2
1-Low, 2-Medium, 3-High									


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MC23CA310	INTERNET OF THINGS	CP	L	T	P	C
		4	2	0	2	3
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand the concepts of IoT and its working models					
2.	To know the various IoT protocols					
3.	To understand about various IoT Physical devices and Endpoints					
4.	To know the security and privacy issues connected with IoT					
5.	To apply the concept of Internet of Things in a real world scenario.					
	INTRODUCTION (Not for Examination)					2
Importance	IOT has become one of the most important technologies of the 21st century. Now that we can connect everyday objects—kitchen appliances, cars, thermostats, baby monitors—to the internet via embedded devices, seamless communication is possible between people, processes, and things.					
Real-life Example(s)	Smart Homes- Healthcare - Transportation					
Linkages	Pre-requisite: Basic programming knowledge, Sensors Future Courses: 5G, blockchain, and edge computing.					
UNIT-I	FUNDAMENTALS OF IOT					6
	Definition and characteristics of IoT, sensors and actuators, physical design of IoT, IoT protocols, IoT communication models, IoT communication APIs, IoT- enabled technologies such as wireless sensor networks, cloud computing, and embedded systems, IoT levels and templates, and domain-specific IoTs including home, city, environment, energy, agriculture, and industry.					
UNIT-II	IOT PROTOCOLS					6
	Protocol standardization for IoT and related efforts, M2M and WSN protocols, SCADA and RFID protocols, issues with IoT standardization, unified data standards, IoT protocols including IEEE 802.15.4, BACnet protocol, Modbus, KNX, Zigbee, network layer, APS layer, and security aspects in IoT protocols.					
UNIT- III	IOT PHYSICAL DEVICES AND ENDPOINTS					6
	Introduction to Arduino and Raspberry Pi, installation and interfaces such as serial, SPI, and I2C, programming fundamentals, and Python programming with Raspberry Pi focusing on interfacing external gadgets, controlling outputs, and reading inputs from GPIO pins.					


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UNIT – IV	INTERNET OF THINGS PRIVACY, SECURITY AND GOVERNANCE	6
	Introduction and overview of governance, privacy, and security issues in IoT, contributions from FP7 projects, security, privacy, and trust in IoT data platforms for smart cities, first steps towards a secure platform, Smartie approach, data aggregation for IoT in smart cities, and security considerations in smart city IoT environments.	
UNIT-V	APPLICATIONS	6
	IoT applications, IoT applications for industry including future factory concepts, brownfield IoT, smart objects, and smart applications, study of existing IoT platforms and middleware, and IoT platforms such as IoT-A and Hydra.	
Total (L)		32 Periods
LIST OF EXPERIMENTS - To study various IoT protocols – 6LowPAN, IPv4/IPv6, Wifi, Bluetooth, and MQTT. - IoT Application Development Using sensors and actuators (temperature sensor, light sensor, infrared sensor) - To study Raspberry Pi development board and to implement LED blinking applications. - To develop an application to send and receive data with Arduino using HTTP request - To develop an application that measures the room temperature and posts the temperature value on the cloud platform. - To develop an application that measures the moisture of soil and post the sensed data over Google Firebase cloud platform. - To develop an application for measuring the distance using ultrasonic sensor and post distance value on Google Cloud IoT platform - Develop a simple application based on sensors. - Develop IoT applications using Django Framework and Firebase/ Bluemix platform. - Develop a commercial IoT application.		
Total (P)		30 Periods
Total (L+P)		62 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy
CO1	Define the infrastructure for supporting IoT deployments	L2 - Understand
CO2	Understand the usage of IoT protocols for communication between various IoT devices	L3 - Apply
CO3	Design portable IoT using Arduino/Raspberry Pi /equivalent boards	L3 - Apply

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CO4	Understand the basic concepts of security and governance as applied to IoT		L3 - Apply	
CO5	Analyze and illustrate applications of IoT in real time scenarios		L4 - Analyze	
	TEXTBOOKS			
1.	Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015, ISBN: 9788173719547.			
2.	Olivier Hersent, David Boswarthick, Omar Elloumi, "The Internet of Things – Key applications and Protocols", Wiley, 2012.			
	REFERENCE BOOKS			
1.	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton, Jerome Henry, "IoT Fundamentals, Networking Technologies, Protocols, and Use cases for the Internet of Things", Cisco Press, First Edition, 2017.			
2.	Dieter Uckelmann, Mark Harrison, Michahelles, Florian (Eds), "Architecting the Internet of Things", Springer, 2011.			
	WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content	
1.	NPTEL – Internet of Things (IIT Kharagpur)	NPTEL – Internet of Things (IIT Kharagpur)	Web References	
2.	Cisco – Introduction to IoT (Free Course)	https://nptel.ac.in/courses/106/105/106105195/	Web References	
3.	oneM2M Official Documentation (Global IoT Standard)	https://skillsforall.com/course/introduction-to-iot	Web References	
	VIDEO REFERENCES			
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Jim Bennett, Senior Cloud Advocate at Microsoft	Lecture	https://www.youtube.com/watch?v=bVffcYh6UBw&utm_source=chatgpt.com
2.	YouTube	Dominik Obermaier	Lecture	https://www.youtube.com/watch?v=i8tU3mDLO-c
3.	YouTube	IBM Technology	Lecture	https://www.youtube.com/watch?v=o5ZsXCVZiiw


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Mapping of COs with POs

COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2		1					1
CO2	3	3	2		1			1	
CO3	2	3	3	2			1		
CO4		2	3	3	2			1	
CO5	3		3	2	3				1
Average	2.8	2.5	2.8	2	2	0	1	1	1

1-Low, 2-Medium, 3-High

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MC23CA601	CAPSTONE PROJECT - I	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To identify a real-world problem and formulate project objectives and scope.					
2.	To design an appropriate solution using programming, algorithms, and modern tools.					
3.	To implement and integrate software modules for a complete system prototype.					
4.	To document the project with proper reports, presentations, and demonstrations.					
5.	To develop teamwork, project management, and professional communication skills					
					Total(P)	60 Periods
Course Outcomes:					BLOOM'S Taxonomy	
Upon completion of this course, the students will be able to:						
CO1	Identify and analyze a real-world problem suitable for software development				L4 - Analyze	
CO2	Design a solution using appropriate algorithms, data structures, and tools				L6 - Create	
CO3	Implement and integrate modules to develop a working prototype				L3 - Apply	
CO4	Prepare project reports and deliver effective presentations				L3 - Apply	
CO5	Demonstrate teamwork, project management, and professional ethics				L3 - Apply	

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	2	2			2		2
CO2	3	2	3	3	2	2	2		2
CO3	2	2	3	3		2	2		2
CO4	2		2	2		3	2		2
CO5	2	2	2	2	2	2	3	3	3
Average	2.2	2.25	2.4	2.4	2	2.75	2.2	3	2.2

1-Low, 2-Medium, 3-High

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MC23PT702	Communication Skills Enhancement - II	CP	L	T	P	C
		2	0	0	2	NC
Programme & Branch	Master of Computer Applications (M.C.A)	Version: 1.0				
Course Objectives:						
1.	To identify main ideas through focused listening.					
2.	To improve oral communication skills for academic and professional purposes.					
3.	To develop the ability to comprehend and interpret varied texts.					
4.	To build skills to develop creative writing.					
	INTRODUCTION (Not for Examination)					2
Importance	Communications skills are crucial for success in campus recruitment. It helps recruiters to understand one's thoughts and ideas clearly during recruitment process. By developing communications, students can get chances of succeeding in campus drive and future professional endeavors.					
Real-life Example(s)	(i) Speaking skills are required for getting hired, advancing in a job, and increasing earning potential. (ii) English communication skills are needed to give an effective presentation.					
Linkages	Communication Skills Enhancement - I					
	LIST OF ACTIVITIES					
	LISTENING					7
1.	Listening to Podcasts/panel discussions and summarizing					
2.	Listening and Note-taking					
3.	Listening to short stories or real-life incidents					
	SPEAKING					8
4.	Narrating real-life incidents or short stories to build fluency					
5.	Small group discussions with a moderator on current issues					
6.	Presenting analysis of a real-world business/technical issues					
7.	Interview skills / Mock interviews (panel, telephonic, video), STAR approach					
	READING					7
8.	Extensive and intensive reading, Cloze tests, and peer correction					
9.	Reading and analyzing industry/academic models (LinkedIn profiles, case studies)					
10.	Reviewing research articles					
	WRITING					
11.	Drafting Minutes of Meeting					

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KOT Campus, Kakani
Salem-637 501

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12.	Creating a LinkedIn profile & blog writing			
13.	Drafting resumes, CVs, and cover letters			
	Total			32 Periods
	Course Outcomes: Upon completion of this course, the students will be able to:			BLOOM'S Taxonomy
CO1	Comprehend the main ideas through focused listening.			L3 - Apply
CO2	Develop clear and confident speaking skills in academic and professional contexts.			L3 - Apply
CO3	Build accuracy in comprehending and interpreting varied texts.			L3 - Apply
CO4	Draft creative and well-structured writing.			L3 - Apply
S.No.	TEXTBOOKS:			
1.	Hudson, C. Communication Skills Training: Master the Art of Effective Interaction and Achieve Success. Barnes & Noble Press, 2024.			
2.	Thilt, John V., and Courtland L. Bovee. Excellence in Business Communication. 14 th edition., Pearson Education, 2024.			
S.No.	REFERENCE BOOKS:			
1.	Sanjay Kumar & Pushp Lata. Communication Skills. 2 nd edition, Oxford Higher Education, 2015.			
2.	Anurag Luharia, Shubham Kadam, Pooja Vyas, and Gaurav Mishra. Modern Approaches in Communication Skills. 1 st edition, Innovative Publication, 2023.			
3.	Dhawan, E. <i>Digital Body Language: How to Build Trust and Connection, No Matter the Distance</i> . St. Martin's Publishing Group, 2021.			
	WEB REFERENCES:			
S.No	Publisher	Website link		Type of Content
1.	Macmillan Education	https://www.macmillanenglish.com/blog-resources/article/communication-skills-resources		variety of resources to develop effective communication
2.	Do2Learn	https://do2learn.com/SocialSkills/CommunicationSkills/index.htm		Communication Skills Resources
	VIDEO REFERENCES:			
S.No.	Video Details	Name of the Expert	Type of Content	Video link
1.	Effective Communication Skills Training Course Video Series	YouTube	The Wizard of Words	https://www.youtube.com/playlist?list=PLq-0mUo3GUDoX6cC-CiXumeCyVhpvXYRkY
2.	Discipline and Success	Sudha Murthy CHAIRPERSON Board of Studies Faculty of CSE & IT Knowledge Institute of Technology	English Speech	https://www.youtube.com/watch?v=vacGRuHDtO0

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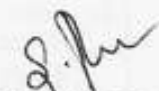
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S.No.	ONLINE COURSES:	
1.	https://www.coursera.org/learn/wharton-communication-skills	
2.	https://online.stanford.edu/courses/gsb-x0011-sharpen-your-communication-skills	

Mapping of COs with POs and PSOs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1						2		1	
CO2						2		1	
CO3						2		1	
CO4						2		1	
Avg.						2.0		1.0	
1-Low, 2 -Medium, 3-High.									


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MC23PT801	APTITUDE SKILLS - I	CP	L	T	P	C
		2	0	0	2	1
Programme & Branch	Master of Computer Applications (M.C.A.)	Version: 1.0				
Course Objectives:						
1.	To know different methods for faster numerical computations					
2.	To learn Logical reasoning skills and improve problem solving abilities.					
3.	To acquire skills required to solve quantitative aptitude problems.					
	INTRODUCTION (Not for Examination)					1
Importance	Problem-solving ability, analytical thinking, and logical reasoning play a vital role in different stages of professional education, career growth, and overall development in the field of computer applications. Therefore, aptitude skills are essential for MCA students in the following areas: 1. Software Design and Development 2. Research and Innovation in Computing 3. Project Planning and Management 4. Competitive Examinations and Career Growth					
Real-life Example(s)	1. Algorithm Optimization During Coding Interviews In technical interviews for IT companies, candidates are required to analyse problems, identify patterns, and develop optimized algorithms within a limited time. Logical reasoning and analytical aptitude support the selection of efficient approaches and help reduce time and space complexity. 2. Database Query Optimization in Software Projects While developing academic or industry-oriented projects, it is necessary to design efficient SQL queries and enhance database performance. Quantitative and analytical aptitude aids in understanding data structures, indexing techniques, and performance optimization. 3. Project Effort Estimation and Resource Planning In software development environments, accurate estimation of timelines, resource allocation, and risk assessment is essential. Numerical aptitude and logical thinking contribute to effective project planning, scheduling, and feasibility analysis.					
Linkages	Aptitude Skills – II					
UNIT-I	SPEED MATHS					6
	Squaring numbers and multiplying numbers faster than the conventional methods-Finding Square roots of numbers faster - Finding Cube roots faster- Solving simultaneous equations faster than conventional methods.					
UNIT-II	LOGICAL REASONING					15
	Alphabet and Number Series - Odd Man Out Series - Puzzles – Blood Relations -Seating Arrangement and Ordering - Directional Sense Test Venn Diagrams – Cubes & Cuboids – Data - Interpretation and Data - Sufficiency .					

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UNIT-III	QUANTITATIVE APTITUDE		9	
	Number Systems - Averages - Percentage -Profit & Loss -Problems on Ages - Ratios & Proportions .			
	Total		31 Periods	
	Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy	
CO1	Apply different techniques for faster calculations .		L3 - Apply	
CO2	Solve mathematical problems by applying logical thinking.		L3 - Apply	
CO3	Exhibit sound knowledge to solve problems of quantitative aptitude.		L3 - Apply	
	REFERENCE BOOKS:			
1.	Aggarwal R. S., "Quantitative Aptitude for Competitive Examinations", S.Chand Publishing Company Ltd(s), 2022.			
2.	Arun Sharma, "How to prepare for Quantitative Aptitude for the CAT" Tata McGraw-Hill Publishing, 2022.			
3.	Praveen R. V., "Quantitative Aptitude and Reasoning" PHI Learning Pvt. Ltd., 2016			
	WEB REFERENCES:			
S.No.	Publisher	Website link	Type of Content	
1.	Indiabix	https://www.indiabix.com/online-test/aptitude-test/	Tests for Practice	
2.	Placement preparation	https://www.placementpreparation.io/quantitative-aptitude/	Tests for Practice	
3.	Geeks for geeks	https://www.geeksforgeeks.org/aptitude-for-placements/	Learning Resources and Tests for Practice	
4.	Prepinsta	https://prepinsta.com/learn-aptitude/	Learning Resources and Tests for Practice	
	VIDEO REFERENCES:			
	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Aptitude buster	Video Lectures	https://www.youtube.com/watch?v=oA0CxMam_Mg&t=864s
2.	YouTube	Freshersworld.com	Video Lectures	https://www.youtube.com/playlist?list=PLJhUHPsqNYkcq6Y0GjwBtTtN7N9

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3.	YouTube	Triumphant Institute of Management Education Pvt. Ltd.	Video Lectures	https://www.youtube.com/@TIME4BankExams/videos
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Mapping of COs with POs and PSOs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2								
CO2	2								
CO3	2								
Avg.	2.0								
1-Low, 2 -Medium, 3-High.									

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UX23PT802	RESEARCH PAPER REVIEW AND PRESENTATION	CP	L	T	P	C
		2	0	0	2	1
Programme & Branch	Common to all M.E (ISE, AE, PED, SE) and M.C.A.	Version : 1.0				

Course Objectives

1. To learn scientific paper reading and wiring skills.
2. To learn the literature review and report writing skills.
3. To understand the research gap and formulation of the research problem.

The work involves the following steps:

- 1 Assigning the faculty supervisor
- 2 Selecting a subject, narrowing the subject into a topic
- 3 Stating an objective.
- 4 Collecting the relevant bibliography (atleast 20 research papers)
- 5 Studying the papers understanding the authors contributions and critically analyzing each paper.
- 6 Preparing a 20-25 page literature review report
- 7 Preparing conclusions based on the literature review report.
- 8 Writing the Final Review Paper
- 9 Final Presentation to the review committee

Evaluation method:

A faculty supervisor will be assigned to each student. The supervisor will assign a topic to the student. The student has to review the literature pertaining to the topic, prepare a 20-25 page report and make a presentation. Minimum 20 research papers have to be reviewed out of which 60% have to be in the recent 05 years. The format for the research paper report and guidelines for assessment will be issued by the Head of the Department before the commencement of the course. The evaluation will be carried out based on the research paper report and presentation, and is evaluated for 100 marks. Minimum 50 marks is essential to pass. In case a student fails, he or she has to redo the course in the forthcoming semesters. Assessment is by Internal Assessment mode only no End Semester Examination.

Semester Examination:		Total(P)	30 PERIODS
COURSE OUTCOMES: Upon completion of this course the students will be able to:			BLOOM'S Taxonomy
CO1	Write a scientific review paper in their field		L3 - Apply
CO2	Identify the research gap and formulate the research problem		L3 - Apply

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1						3			3
CO2		3							3
Average		3				3			3
1-Low, 2-Medium, 3-High									

PROFESSIONAL ELECTIVE VERTICAL I : JAVA FULL – STACK DEVELOPMENT

MC23CA401	WEB DEVELOPMENT FUNDAMENTALS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand the fundamentals of Computer Networks and the Internet.					
2.	To design web pages with HTML and XML.					
3.	To develop visual presentation and responsive web pages using CSS.					
4.	To create an interactive web page using advanced JavaScript.					
5.	To develop responsive web layouts using bootstrap and work with jQuery for dynamic web interactions.					
	INTRODUCTION (Not for Examination)					2
Importance	- Understanding the basics of HTML, CSS, and JavaScript provides a solid foundation for learning more advanced topics and frameworks. - Mastering the fundamentals helps you develop problem-solving skills essential for tackling more complex development challenges. - Familiarity with web development fundamentals helps you adhere to web standards and best practices, ensuring compatibility and accessibility. - Creating User-Centric Designs: User Experience (UX) - Learning how design affects usability. Responsive Design - Ensuring websites work across various devices and screen sizes.					
Real Life Example(s)	Building an Event Booking Website: Developing a web application for a local event management company. The website will allow users to browse upcoming events, book tickets, and receive confirmation emails.					
Linkages	Pre-requisite: Computer Networks, Problem Solving using C Programming, Python for Data Science Future Courses: C# and .NET					
UNIT-I	INTRODUCTION TO WEB DESIGN					9
	Computer Network Basics – Introduction to Internet – History – WWW – Evolution of Web – Web Browser – URI – Web Server – Website – Web System Architecture (Single Tier, Two Tier, Three Tier Architecture) – HTTP Request and Response – Port – DNS – Introduction to Client-side and Server-side scripting. ❖ Use Figma to design a "User Feedback Form" with fields for name, email, comments, and a submit button, ensuring a user-centered design.					


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UNIT-II	HYPER TEXT MARK-UP LANGUAGE & EXTENSIBLE MARKUP LANGUAGE	9
	Basic HTML Elements – Attributes – Hyperlinks and bookmarks – Images – Favicon – Emojis – List and Table – Form Elements – Layouts – Other HTML Elements – Inline Vs Block level elements – Media and Graphics. ❖ Create a feedback form with a <h2> titled "Feedback Form," a text input for the name, an email input for contact, a text area for comments, and a submit button	
UNIT-III	CASCADING STYLE SHEET	9
	Basics of CSS – Types – Selectors – Pseudo classes – Core CSS3 Properties – CSS3 Features – Media Queries CSS Preprocessors (SASS, SCSS). ❖ Create a simple webpage with a product card that includes: Product Card, Buy Now Button.	
UNIT-IV	JAVASCRIPT BASICS	9
	Introduction to JavaScript – HTML Vs CSS Vs JavaScript – Syntax – First JavaScript – Keywords) – Variables – Operators – Data Types – Type Conversion – Control Flow Statements – Functions – Event listener – Constructor – Prototypes – Objects: String, Array, Math, Date, Regular Expressions, Events – DOM – Form Validation – Canvas. ❖ Create user validation in Registration Form.	
UNIT-V	BOOTSTRAP AND JQUERY	9
	Basics of Bootstrap – Grid System and Layout – Navbar and Navigation Components – Buttons and Alerts – Forms and Input Groups – Cards and Modals – Create Responsive Web Page – Introduction to jQuery – Selecting and Manipulating DOM Elements – Event Handling – jQuery Plugins ❖ Create a 3-column grid with cards that include a title, image, and description.	
	Total(L)	47 Periods
	❖ Internal Evaluation Only	
	OPEN ENDED PROBLEMS/QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	


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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy	
CO1	Understand the concepts of Computer Networks and the Internet.	L2 - Understand	
CO2	Develop a web page using HTML elements, forms, tables and XML	L3 - Apply	
CO3	Design responsive, visually appealing, and maintainable web pages using CSS.	L3 - Apply	
CO4	Apply Javascript objects to build dynamic web page with validation.	L3 - Apply	
CO5	Develop a responsive web page using Bootstrap's and integrate jQuery plugins to enhance website features.	L3 - Apply	
TEXTBOOKS			
1.	Philip Ackermann, Full-Stack Web Development: The Comprehensive Guide, Rheinwek Computing, First Edition, 2023.		
2.	John Dean, "WEB PROGRAMMING with HTM, CSS and JavaScript", First Edition, Jones & Bartlett Learning, 2018.		
3.	Joe Fawcett, Danny Ayers, Liam R. E. Quin, <i>Beginning XML</i> , Wrox Publications, 5th Edition, 2022.		
REFERENCE BOOKS			
1.	Nabendu Biwas, Ultimate Full-Stack Web Development with MERN, Orange Education Pvt Ltd., 2023.		
2.	Jennifer Robbins, "Learning Web Design: A Beginner's Guide To HTML, CSS, JavaScript, And Web Graphics", Fifth Edition, O'Reilly Publisher, 2018.		
3.	A. Powell, "HTML & CSS: The Complete Reference Thomas" Fifth Edition, Tata McGraw Hill, 2010.		
4.	Lea Verou, "CSS Secrets: Better Solutions to Everyday Web Design Problems", First Edition, O'Reilly Publisher, 2015.		
WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content
1.	Freecodecamp	https://shorturl.at/aa5aY	Articles
2.	Web.dev	https://shorturl.at/IntBl	Articles
3.	w3schools	https://shorturl.at/flfgp	Others
4.	Hubspot	https://shorturl.at/7kCi6	Others
5.	Valuecoders	https://shorturl.at/eNffI	Articles


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VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video link
1.	YouTube (NPTEL)	Madhavan Mukund	Lecture	https://shorturl.at/3abn1
2.	YouTube	Zach Gollwitzer	Lecture	https://shorturl.at/S1sJj
3.	YouTube	Zach Gollwitzer	Lecture	https://shorturl.at/4hcUT

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	2						2	2
CO2	2		3	2		2			
CO3			3	2		2			
CO4	2	2	3	3					
CO5	2		3	3		2	2		2
Average	2	2	3	2.5		2	2	2	2
1-Low, 2-Medium, 3-High									




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MC23CA402	FRONT-END DEVELOPMENT	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To develop interactive web solutions by applying ES6 features in JavaScript.					
2.	To apply UI/UX design principles and techniques to build user- friendly interfaces.					
3.	To gain knowledge in building web applications using React.					
4.	To apply state management concepts by implementing Redux and Redux Toolkit in web applications.					
5.	To build websites using Next.js for search engines optimization.					
	INTRODUCTION (Not for Examination)					2
Importance	Front-End technologies develop the user interface and interactive elements of websites. Enhancing user experience is achieved by enabling intuitive navigation and engagement. Compatibility across various devices and browsers is ensured through front-end solutions.					
Real-life Example(s)	Designing and developing E-commerce platforms and Dashboards for online learning platform.					
Linkages	Pre-requisite: Fundamentals of Web development Future courses: Back-End Development, DevOps and Cloud basics.					
UNIT-I	ADVANCED JAVASCRIPT					9
	Overview of Core JavaScript Features – ES6+: Let and const scope – Arrow functions – Template literals – Destructuring – Spread and Rest operators – Default parameters. Functions and Callbacks: Higher-order functions – Callback functions – Closures-Anonymous and named functions – Function scope vs block scope. Advanced Array Methods: map – filter – reduce. Asynchronous JavaScript: set Timeout – set Interval Event loop and call stack basics – Callbacks vs Promises – Creating and Chaining Promises – async/await, Error handling in Promises. Working with APIs: fetch API – Handling JSON responses. Introduction to JavaScript Frameworks and Libraries.					
UNIT-II	UI/UX DESIGN					9
	Overview of Design Thinking – Introduction to UI/UX: Importance of User Interface and User Experience – UI vs UX: Key Differences – Fundamental Principles of Good Design – Figma Tool Overview: Introduction to Figma and Interface Walkthrough – Creating Mindmaps – Designing Wireframes: Low-Fidelity – Mid-Fidelity – High-Fidelity – Building Mockups and Interactive Prototypes.					


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UNIT-III	REACT JS	9
	Introduction to React – Virtual DOM – Setting up the React Development Environment – Components: Functional Class – Props – State Management – Conditional Rendering – Event Handling in JSX – Controlled vs Uncontrolled Components – Form Submission – React Hooks: use State Hook – use Effect Hook – use Context Hook – Lifecycle simulation with Hooks – Rules of Hooks – Fetch API – Axios – Handling loading and error states – React Router Basics – Route – Link – Navigation – Dynamic Routing.	
UNIT-IV	REDUX	9
	Introduction: Redux – Need for State Management – Store – Actions – Reducers – Setting up Redux in React App – Redux Flow – Provider Component – useSelector Hook – useDispatch Hook – Connecting Redux to Components – Middleware & Async Handling: Redux Middleware – Redux Thunk – Async Actions – Handling API Calls with Redux.	
UNIT-V	INTRODUCTION TO NEXT.JS AND PROJECT DEVELOPMENT	9
	Next.js: Differences from React – File-based Routing – Pages and Navigation – Static and Server-side Rendering – API Routes – Data Fetching (getStaticProps, getServerSideProps) – Styling in Next.js – SEO with Head component – Image Optimization. ❖ Project Development: Final Mini Project using React – Redux and Next.js – Deploy project on Vercel.	
	Total(L)	47 Periods
	❖ Internal Evaluation Only	
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy
CO1	Develop interactive web solutions by applying ES6 features in JavaScript.	L3 - Apply
CO2	Apply UI/UX design principles and techniques to create user-friendly interfaces.	L3 - Apply
CO3	Demonstrate React concepts in creating web applications.	L3 - Apply
CO4	Apply state management concepts through implementation of Redux and Redux Toolkit in web applications.	L3 - Apply
CO5	Build SEO-friendly websites using Next.js to enhance search engine visibility.	L3 - Apply

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	TEXTBOOKS			
1.	Angel Sola Orbaiceta, "Build a Front -End Web Framework (From Scratch)", 1 st Edition, Manning Publications, 2024.			
2.	Kirupa Chinnathambi, "Learning React: A Hands-On Guide to Building Web Applications Using React and Redux", 2 nd Edition, Addison-Wesley Professional, 2018.			
3	Canva Team, "Designing with Canva: A Practical Guide for Beginners", Canva Press, 2024.			
	REFERENCE BOOKS			
1.	Joel Marsh, "UX for Beginners: A Crash Course in 100 Short Lessons", 1 st Edition, O'Reilly, 2022.			
2.	Alex Banks, Eve Porcello, "Learning React: Functional Web Development with React and Redux", 1 st Edition, O'Reilly, 2020.			
3.	Shama Hoque, "Full-Stack React Projects", 2 nd Edition, Packo Publishing Ltd., 2020.			
4.	Canva Design School, "Canva for Visual Communication", Canva Publications, 2023.			
	WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content	
1.	Free Code Camp	https://www.freecodecamp.org/learn/front-end-development-libraries/#sass	Web Reference	
2.	W3schools	https://www.w3schools.com/js/DEFAULT.asp	Web Reference	
3.	Interaction Design Blockchain (IxDF)	https://www.interaction-design.org/literature	Web Reference	
	VIDEO REFERENCES			
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Mr.Prashant	Tutorial	https://youtu.be/eILUmCJh4?si=uExYncWm-CQsg7Ud
2.	YouTube	Mr.Mosh	Lecture	https://youtu.be/poQXNp9It?si=0qB81_2h1mz7xXym
3.	YouTube	Codevolution	Tutorial	https://www.youtube.com/playlist?list=PLC3y8-rFHvwhIEc4I4YsRz5C7GOBnxSJY


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Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	2	3	3					2
CO2		2	2	2		3		2	2
CO3		2	3	3					2
CO4		2	3	3					2
CO5		2	3	3		2		2	3
Average	2	2	2.8	2.8		2.5		2	2.2
1-Low, 2-Medium, 3-High									


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MC23CA407	WEB DEVELOPMENT FUNDAMENTALS LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				

Course Objectives

1.	To introduce the structure of HTML documents and familiarize students with essential elements, tags, layouts, forms, tables, and controls.
2.	To enable students to apply basic and advanced text formatting techniques for designing structured and readable web pages.
3.	To develop skills in creating and managing style sheets using CSS, including properties, box model, positioning, and responsive design concepts.
4.	To train students in designing visually appealing web pages using internal and external CSS for effective styling and layout management.
5.	To introduce JavaScript fundamentals and guide students in developing simple interactive web applications using scripts.

LIST OF EXPERIMENTS

1. Basic Web Page Development

Design and develop a basic webpage titled **"My First Web Page"** that displays a heading, a welcome paragraph, a horizontal rule, and contact information using HTML structural tags. The page must follow correct HTML document structure.

2. Student Registration Form

Create a **Student Registration Form** in HTML that collects Name, Email ID, Gender, Date of Birth, Skills, Department, Contact Number, and Address. The form should include radio buttons, checkboxes, dropdown lists, validation attributes, and a submit button.

3. Navigation Page using Hyperlinks, Lists and Tables

Develop a mini website that contains **three linked pages**: Home, Courses and Contact pages. Use ordered/unordered lists for navigation items and use a formatted table in the Courses page to display course offerings.

4. Online Gallery using HTM Media Elements

Build an HTM webpage that displays an image gallery of five images, embeds a short video, and plays an audio clip. Include captions and adjust size and alignment using attributes.

5. CSS-Based UI Styling

Apply CSS styles to a webpage by using selectors (class, id and pseudo selectors) to style headings, paragraphs, navigation links, and buttons. The output should show hover effects and changes in font, color, margin and padding.

6. Web Page Layout using CSS Flexbox/Grid

Design a **3-section webpage layout** using CSS Flexbox/Grid containing a header (with title), main content, and footer area. Ensure proper alignment, spacing, and responsive behavior.

7. Responsive Product Showcase using Media Queries

Create a **product showcase webpage** using media queries. Display 4 product cards in desktop view, 2 cards in tablet view, and 1 card in mobile view. All layout changes must happen using media queries.

8. JavaScript DOM Manipulation

Write a webpage that allows the user to dynamically **add, delete and update list items** using JavaScript DOM methods. Use buttons and input fields to control actions.

9. JavaScript Client-Side Form Validation

Develop a webpage with a **registration/login form** and validate fields using JavaScript (name length check, email format check, password strength check and number validation). Display error messages dynamically without page refresh.

10. Experiment 10 – Bootstrap Based Responsive Portfolio Website

Build a responsive portfolio website using Bootstrap. Include navigation bar, jumbotron/banner, card-based project section, contact form, and footer. Ensure layout adapts to different screen devices using Bootstrap grid.

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Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15.12.2025

HARDWARE/SOFTWARE REQUIREMENTS**Hardware:**

i3 Processor, 4 GB RAM, 250 GB HDD/SSD, 17" Monitor, Keyboard, Mouse, Internet connection, UPS.

Software:

Windows/Linux OS, Chrome/Firefox browser, Visual Studio Code/Notepad++, HTM, CSS3, JavaScript ES6.

Total(P)**60 Periods****Course Outcomes:****Upon completion of this course, the students will be able to:****BLOOM'S
Taxonomy**

CO1	Understand the fundamental concepts of HTML elements, tags, structure, and text formatting for creating basic web pages.	L2 - Understand
CO2	Design structured web pages using HTML layouts, forms, lists, frames, tables, and various control elements.	L3 - Apply
CO3	Apply CSS properties, selectors, box model, and positioning techniques to style and enhance web page presentation.	L3 - Apply
CO4	Develop well-designed web interfaces using internal and external style sheets for professional and responsive layouts.	L4 - Analyze
CO5	Implement JavaScript to add interactivity and build simple dynamic web applications.	L3 - Apply

Mapping of COs with POs

COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	1	2		2				
CO2	2		3	2	2			2	
CO3	2		3	2	3			2	
CO4	2	2	3	3	3			2	
CO5	2		3	3	2			2	2
Average	2	1.5	2.8	2.5	2.4			2	2

1-Low, 2-Medium, 3-High


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MC23CA408	FRONT-END DEVELOPMENT LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				

Course Objectives

1.	To enable students to design user-centric interfaces through wire framing and prototyping using modern UI/UX tools.
2.	To develop competence in building interactive and dynamic web applications using JavaScript and ES6 concepts.
3.	To facilitate understanding of component-based development using React for creating scalable web applications.
4.	To apply state management concepts using Redux and Redux Toolkit in real-time applications
5.	To equip students with skills to develop SEO-friendly, fast-loading web applications using Next.js and modern routing techniques.

LIST OF EXPERIMENTS

1. ES6 Advanced JavaScript Practice

Write a JavaScript program that uses ES6 features (let/const, arrow functions, template literals, spread/rest operator and destructuring) to calculate the average score of students from an array of student objects, and display formatted results in the browser.

2. Asynchronous Programming & API Fetch

Develop a web page that fetches real-time data from a public API (such as posts or users) using fetch, handles JSON output, and displays results in a structured list. Use promises or async/await with error handling.

3. UI/UX Wireframe & Prototype using Figma

Using Figma, design a **High-Fidelity UI Prototype** for a "Movie Ticket Booking Portal" that includes a home page, movie listing screen, seat selection screen and payment screen. Produce a clickable interactive workflow.

4. Functional React Component & Props

Create a React application that displays a list of product cards using functional components. Pass product details as props to child components and render them dynamically.

5. React Hooks & Form Handling

Build a **User Signup Page** in React that uses useState to manage form data and useEffect to print state changes in console. Implement form validations and show errors without page reload.

6. React Routing and Dynamic Navigation

Develop a multipage React application using React Router containing pages: Home, About, Products and Product Details. Implement dynamic routing to open individual product detail pages using product IDs.

7. Redux State Management

Create a React-Redux based counter application where:

Redux store holds state values

Actions increment and decrement count

useSelector retrieves state

useDispatch triggers actions

Display live updates on the screen.

8. Redux Async Operations using Middleware

Implement Redux Thunk to fetch data from an API (e.g., list of blog posts), store it in Redux store and display the results on a React page. Include loading and error message states.

9. Next.js Routing & Data Fetching

Develop a Next.js application with multiple pages using file-based routing:

Home page

About page


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Blogs page (data fetched using `getStaticProps`)
 Display blog titles as navigation links to individual Blog Details pages.

10. Front-End Mini Project Development

Build a complete front-end project using **React + Redux OR Next.js** that solves a real-world problem (e.g., Todo App, Weather Dashboard, Student Portal, Ecommerce Catalog). Project must include:

- UI/UX design blueprint
- components, hooks and routing
- Redux or Context state handling
- API integration
- responsive UI
- Deploy final project using Vercel.

HARDWARE/SOFTWARE REQUIREMENTS

Hardware Requirements

- Desktop/Laptop: Intel i3 or higher, 8 GB RAM, 10 GB free disk
- Internet connection (for package installation & API usage)

Software Requirements

- OS: Windows / Linux / macOS
- Node.js (LTS version)
- Code Editor: Visual Studio Code
- Browsers: Chrome / Firefox
- Figma (Web version)
- Frameworks/Libraries: React, Redux, Redux Toolkit, Next.js, JavaScript ES6

		Total(P)	60 Periods
Course Outcomes:			BLOOM'S Taxonomy
Upon completion of this course, the students will be able to:			
CO1	Build interactive applications using JavaScript to perform data processing, validation, and dynamic updates.		L3 - Apply
CO2	Apply UI/UX design principles to create mid/high-fidelity wireframes and interactive prototypes using Figma.		L3 - Apply
CO3	Develop responsive and scalable web applications using React components, routing, and API integration.		L3 - Apply
CO4	Implement global state management in complex applications using Redux and Redux Toolkit.		L4 - Analyze
CO5	Create SEO-friendly and fast-rendering websites using Next.js with file-based and dynamic routing.		L3 - Apply

Mapping of COs with POs

COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2	3	3					2
CO2	2	2	3	3		2	2	2	2
CO3	2	2	3	3	2		2		2
CO4	2	2	3	3	2		2		2
CO5	2	2	3	3	2		2		3
Average	2.6	2	3	3	2	2	2	2	2.2

1-Low, 2-Medium, 3-High

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 Approved in Academic Council Meeting held on 15.12.2025

Vertical : II AUTOMATION TESTING

MC23CA413	TESTING FUNDAMENTALS AND MANUAL TESTING	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand the fundamentals, need and process of software testing.					
2.	To learn various SDLC models and software testing methodologies.					
3.	To apply test case design techniques and prepare necessary documentation.					
4.	To understand bug life cycle and explore test management tools and techniques.					
5.	To apply agile testing principles and manage test projects using tools like Jira.					
	INTRODUCTION (Not for Examination)					2
Importance	- Ensures delivery of quality software by identifying and resolving defects early. - Builds industry-ready skills in testing, essential for software development and QA roles. - Strengthens knowledge of SDLC models, test strategies, and defect tracking-a core part of every software project. - Encourages analytical and critical thinking through test design techniques and reporting. - Provides hands-on exposure to tools like Jira, aligning with Agile industry practices.					
Real-life Example(s)	Online Banking System, Railway Booking App, E-commerce App.					
Linkages	Pre-requisite: Python Programming. Future courses: Automation Testing with Selenium, TestNG, Performance and Database Testing.					
UNIT-I	INTRODUCTION TO SOFTWARE TESTING					9
	Software Testing Introduction – Importance of Software Testing – what is Testing? – Errors, Bugs, Defects, Failures – Principles of Software Testing – SDLC Phases – Role of Testing in SDLC – Overview of Verification and Validation – Introduction to STLC.					
UNIT-II	SDLC MODELS AND TESTING METHODOLOGIES					9
	Waterfall Model – V-Model – Agile Model – Iterative Model – Spiral Model – Big Bang Model – Introduction to Software Testing Methodologies – White Box Testing – Black Box Testing – Grey Box Testing – Static vs Dynamic Testing – Comparison of SDLC Models.					


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UNIT-III	TEST CASE DESIGN TECHNIQUES AND DOCUMENTATION	9
	Test Case vs Test Scenario – Test Case Design Techniques: Boundary Value Analysis – Equivalence Partitioning – Decision Table – State Transition – Use Case Testing – Requirement Traceability Matrix (RTM) – Test Plan Preparation – Entry and Exit Criteria – Test Case Template.	
UNIT-IV	BUG LIFE CYCLE AND TEST MANAGEMENT	9
	Defect/Bug Life Cycle – Severity and Priority – Bug Report Preparation – Logging Bugs with Screenshots – Different Test Suites and Types: Smoke, Sanity, Regression, Retesting, Alpha and Beta Testing – Overview of QA and QC – Roles in Testing Team – Communication with Developers.	
UNIT-V	AGILE TESTING AND TESTING PROJECT	9
	Agile Methodology Overview – Scrum Roles and Ceremonies – Writing User Stories and Acceptance Criteria – Introduction to Jira for Test Management – Logging Test Cases and Bugs – Sprint Board and Backlog Management. Testing Project: Writing Effective Test Cases – Defect Reports – Traceability Matrix – Hands-on with Test Management in Jira.	
	Total(L)	47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain software testing concepts, principles, and its role in SDLC.	L2 - Understand
CO2	Differentiate between testing methods and SDLC models.	L2 - Understand
CO3	Apply different design test cases using standard techniques and document testing artifacts.	L3 - Apply
CO4	Manage defects using bug life cycle concepts and differentiate various testing types.	L3 - Apply
CO5	Implement Agile testing concepts and use Jira for test case management in real-time projects.	L3 - Apply


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	TEXT BOOKS			
1.	Srinivasan Desikan and Gopalaswamy Ramesh, "Software Testing: Principles and Practices", 2 nd Edition, Pearson Education, 2005.			
2.	Dorothy Graham, Rex Black, Erik van Veenendaal, "Foundations of Software Testing: ISTQB Certification", 4 th Edition, Cengage Learning, 2020.			
	REFERENCE BOOKS			
1.	Ron Patton, "Software Testing", 2 nd Edition, Sams Publishing, 2005.			
2.	William E. Perry, "Effective Methods for Software Testing", 3 rd Edition, Wiley, 2006.			
3.	Paul C. Jorgensen, "Software Testing: A Craftsman's Approach", 4 th Edition, CRC Press, 2013.			
	WEB REFERENCES			
S. No.	Publisher	Website link		Type of Content
1.	Guru99	https://www.guru99.com/software-testing.html		Course Content
2.	Atlassian	https://www.atlassian.com/software/jira/guides		Web Content
	VIDEO REFERENCES			
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Simplilearn	Lecture	https://www.youtube.com/live/6EMg7-li0y4?si=EyZmr5pw6D6Oi6II
2.	YouTube	Edureka	Lecture	https://youtu.be/QPU3_JGrtU?si=jZHY2Si1bWRU_cS9

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2							2
CO2	2	3							2
CO3	2	2	3	2					2
CO4	2	2	3	2			2		2
CO5	2	2	3	3	2	2	2		2
Average	2.2	2.6	3	2.3	2	2	2		2
1-Low, 2-Medium, 3-High									


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MC23CA414	AUTOMATION TESTING WITH SELENIUM	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand the fundamentals of automation testing and compare it with manual testing to determine suitable testing scenarios.					
2.	To explore Selenium and its components to identify and interact with various web elements using locators like XPath, CSS Selectors.					
3.	To implement browser-based automation using Selenium WebDriver to handle user actions, alerts, frames, and multiple browser windows.					
4.	To apply data-driven testing techniques using Excel/flat files and manage synchronization issues using appropriate wait strategies.					
5.	To design structured test automation frameworks using Page Object Model (POM), incorporate logging and error handling mechanisms for real-time testing.					
	INTRODUCTION (Not for Examination)					2
Importance	Automation Testing with Selenium is a critical skill in the software development lifecycle. As industries shift toward agile and DevOps practices, there is a growing demand for faster and more reliable testing methods. This course equips students with the knowledge and hands-on experience to automate web application testing using Selenium, one of the most widely used open-source automation tools in the software industry.					
Real-life Example(s)	E-commerce Testing, Banking Applications, Social Media Platforms.					
Linkages	Pre-requisite: Java Programming, Fundamentals of Web Development, Object Oriented Software Engineering. Future courses: Advanced Test Testing with TestNG and CI/CD, Tools for Automation Testing, Performance Testing and Database Testing.					
UNIT-I	INTRODUCTION TO AUTOMATION TESTING AND SELENIUM					9
	Introduction to Automation Testing – Manual Testing vs Automation Testing – When to go for Automation? – Overview of different Automation Tools – Advantages and Limitations of Automation Testing – Introduction to Selenium – Features of Selenium – Selenium Components and Architecture – Installing Selenium and Configuring Eclipse IDE.					
UNIT-II	WEB ELEMENTS AND LOCATORS IN SELENIUM					9
	Basics of HTML Web Elements – Types of Selenium Locators: id, className, name, linkText, partialLinkText, tagName, CSS Selector, XPath – Writing and using XPath effectively – Introduction to SelectorsHub – Inspecting and locating UI elements – Handling Dynamic Elements.					


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UNIT-III	SELENIUM WEBDRIVER AND USER ACTIONS	9
	Launching browsers using WebDriver – Working with web elements: click, sendKeys, getText, isDisplayed, clear – Handling alerts and popups – Handling frames and nested frames – Switching between multiple windows and tabs – Mouse hover and keyboard operations using Actions class – Handling Tooltips and Drag-and-Drop actions.	
UNIT-IV	DATA-DRIVEN TESTING AND SYNCHRONIZATION	9
	What is Data-Driven Testing – Creating data-driven tests using Excel and Flat files – Apache POI Basics – Synchronization in Selenium: Implicit Wait, Explicit Wait, Fluent Wait – Managing page load times – Taking screenshots on failures – Handling cookies and JavaScript popups.	
UNIT-V	SELENIUM FRAMEWORK AND REAL-TIME TESTING	9
	Introduction to Selenium Frameworks – Page Object Model (POM) – Page Factory implementation – Data-driven testing using Excel and Flat Files – Screenshot capture on failure – Handling Tooltips and Drag-and-Drop elements – Config Properties – Log4j logging – Handling Cookies.	
	Total(L)	47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the concepts of automation testing, differentiate it from manual testing, and identify appropriate scenarios for automation.	L2 - Understand
CO2	Identify and apply various Selenium locators and WebDriver commands to interact with web elements.	L3 - Apply
CO3	Demonstrate the ability to handle frames, alerts, windows, and perform advanced user interactions.	L3 - Apply
CO4	Develop data-driven automated test scripts using Excel/flat files and implement synchronization strategies.	L3 - Apply
CO5	Design and implement maintainable test automation frameworks using POM, Page Factory, and integrate logging.	L3 - Apply
	TEXT BOOKS	
1.	Unmesh Gundecha, "Selenium Testing Tools Cookbook", 2 nd Edition, Ingram short title, 2015.	
2.	Unmesh Gundecha and Satya Avasaral, "Selenium WebDriver 3 Practical Guide", 2 nd Edition, Packt Publishing Ltd, 2018.	

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	REFERENCE BOOKS			
1.	Navneesh Garg, "Test Automation using Selenium WebDriver with Java", Atlantic Publishers and Distributors (P) Ltd, 2014.			
2.	Varun Menon, "TestNG Beginner's Guide", Packt Publishing Limited, 2013.			
3.	Unmesh Gundecha, "Learning Selenium Testing Tools with Python", Packt Publishing Limited, 2014.			
	WEB REFERENCES			
S. No.	Publisher	Website link		Type of Content
1.	Selenium Official	https://www.selenium.dev/documentation/		Learning Content
2.	Guru99	https://www.guru99.com/selenium-tutorial.html		Web Content
3.	ToolsQA	https://www.toolsqa.com/		Course Content
	VIDEO REFERENCES			
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Mukesh Otwani	Lecture	https://youtu.be/ft8L9zVJOFY?si=-Ec29kC4vCwh3s_A
2.	YouTube	Rahul Shetty	Demonstration	https://youtu.be/BHFHGgfIh0s?si=joUdas_3T_aUaENx

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2							2
CO2	2	3							2
CO3	2	2	3	2					2
CO4	2	2	3	2			2		2
CO5	2	2	3	3	2	2	2		2
Average	2.2	2.6	3	2.3	2	2	2		2
1-Low, 2-Medium, 3-High									


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MC23CA419	TESTING FUNDAMENTALS AND MANUAL TESTING LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				

Course Objectives

1.	To understand the fundamental concepts of software testing, its types, and role in SDLC.
2.	To differentiate between SDLC models and testing methods for effective planning.
3.	To apply standard techniques to design test cases and document testing artifacts.
4.	To manage and classify defects using bug life cycle concepts and reporting tools.
5.	To implement Agile testing practices and utilize tools like Jira for test management.

LIST OF EXPERIMENTS

1. STLC Documentation for Simple Application

Study the Software Testing Life Cycle (STLC) and prepare a structured **testing workflow document** for a simple Login module, including requirement analysis, test planning, test execution, defect logging, and test closure stages.

2. SDLC Model Identification & Comparison

Analyze a given real-time project scenario (e.g., Hospital Management System) and **identify a suitable SDLC model**. Justify the selection and compare the model against at least two other SDLC models based on features, timeline, risk handling, and customer involvement.

3. Black Box Test Case Development

Using Black Box Testing techniques, create **detailed Test Scenarios and Test Cases** for a User Login form that includes fields for username, password, login button, and forgot password link. Include test data, expected results, priority, preconditions, and postconditions.

4. Boundary Value & Equivalence Partitioning Test Design

For an online shopping cart's quantity input (range 1-20), design **test cases using Boundary Value Analysis and Equivalence Partitioning** to validate correct and incorrect selections. Present the cases in standard template form.

5. State Transition & Decision Table Testing

Create a **state transition table and decision table test cases** for an ATM PIN validation system. Include scenarios like correct PIN, incorrect PIN, multiple failures, system lock state, and reset state.

6. Requirement Traceability Matrix Creation

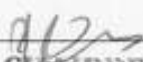
Prepare a **Requirement Traceability Matrix** linking the features of a library management system to their respective test cases and expected outcomes to ensure 100% requirement coverage.

7. Bug Life Cycle Demonstration

Simulate the **bug life cycle** for a search feature defect in an e-commerce application. Document every defect stage (New → Assigned → Open → Re-test → Verified → Closed) including severity, priority and supporting screenshots.

8. Test Suite & Test Execution Report

Design and implement **Smoke, Sanity and Regression Test Suites** for a Ticket Booking web module. Execute the tests manually and submit a structured test execution report with status, test evidence, and retest logs.


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9. Agile User Story Based Testing

Write Agile-style **User Stories and Acceptance Criteria** for a Hotel Booking application and convert them into test cases. Present execution results in the form of Sprint Testing Report.

10. Jira Test Management & Bug Logging

Using Jira perform the following:

- create a test project,
- upload test cases,
- execute tests,
- log a defect with severity & priority,
- track the sprint board status.

Generate and submit reports from Jira dashboard.

HARDWARE/SOFTWARE REQUIREMENTS**Hardware Requirements**

- Desktop/Laptop with **Intel i3 or higher, 4–8 GB RAM**
- **10 GB free disk space**
- Internet connection (for Jira and web resources)

Software Requirements

- OS: Windows / Linux / macOS
- Browser: Google Chrome / Firefox
- Jira account (free cloud version sufficient)
- IDE/Editor: Visual Studio Code / Eclipse / any code editor for sample code testing

		Total(P)	60 Periods
Course Outcomes:			BLOOM'S Taxonomy
Upon completion of this course, the students will be able to:			
CO1	Explain software testing concepts, types of defects, and their role in SDLC		L2 - Understand
CO2	Differentiate SDLC models and testing methodologies		L2 - Understand
CO3	Apply techniques like Boundary Value Analysis and Equivalence Partitioning to design test cases		L3 - Apply
CO4	Manage defects and classify bugs based on severity and priority		L3- Apply
CO5	Use Agile testing practices and manage test cases/bugs using Jira		L3 - Apply

Mapping of COs with POs

COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2							2
CO2	2	3							2
CO3	2	2	3	2					2
CO4	2	2	3	2			2		2
CO5	2	2	3	3	2	2	2		2
Average	2.2	2.6	3	2.3	2	2	2		2
1-Low, 2-Medium, 3-High									

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w.e.f. 22.12.2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15.12.2025

MC23CA420	AUTOMATION TESTING WITH SELENIUM LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				

Course Objectives

1.	To understand Selenium installation, configuration, and its components for automation testing
2.	To identify and interact with web elements using different locators
3.	To handle frames, alerts, windows, and perform advanced user interactions
4.	To implement data-driven testing using Excel/flat files and apply synchronization techniques
5.	To design and implement maintainable test automation frameworks using POM/Page Factory and integrate logging

LIST OF EXPERIMENTS

1. Selenium WebDriver Setup & First Automation Script

Install Selenium WebDriver and configure the Eclipse IDE. Develop a simple script to launch a browser, open Google home page, verify the title, and close the browser. Demonstrate automation flow and document output.

2. Locator Identification & Web Element Interaction

Automate a Login form of a sample web application. Locate each field using different Selenium locators (id, name, XPath, CSS selectors, linkText, tagName) and perform actions like enter text, click, clear, and read element text.

3. XPath & CSS Selector Automation

Automate an e-commerce product search scenario:

- search item through search bar,
 - apply filters using XPath axes and dynamic attributes,
 - retrieve and print product names and pricing.
- Demonstrate both CSS and XPath strategies.

4. Handling Alerts, Frames, & Windows

Develop a Selenium script to automate an Online Form submission page containing:

- JavaScript alerts,
 - nested frames,
 - multiple browser tabs/windows.
- Switch between them and validate element presence.

5. Mouse & Keyboard Actions Using Action Class

Automate a webpage to:

- perform mouse hover menu navigation,
 - execute drag-and-drop,
 - right-click and double-click interactions,
 - simulate keyboard inputs.
- Verify visible state changes after each action.

6. Data-Driven Automation Testing using Excel

Develop an automated Login Test script using Apache POI to read multiple sets of input data (username & password) from an Excel sheet. Validate login success/failure and store results back into Excel.

7. Synchronization & Wait Implementation

Automate a web page that contains dynamically loading content. Implement implicit wait, explicit wait, and fluent wait to synchronize interactions, ensuring element stability before clicking or sending input

8. Handling Cookies, Screenshots & Page Load Issues

Create a Selenium script that:

- navigates to a web application,
- prints cookies, deletes specific cookies,
- captures screenshots upon failure and success,
- controls page load timeout settings.

9. Page Object Model (POM) Framework Implementation

Design a Selenium POM-based framework to automate Login and Logout functionality. Implement page classes, reusable functions, constructors, and page navigation without repeated code.

10. Mini Automation Project using Selenium

Develop a complete automation testing mini project for any chosen application (e-commerce, student registration portal, banking UI) Include:

- Test scenarios
- Test scripts
- Data-driven execution
- reusable functions
- logging
- screenshots
- folder structure

Deliver final execution report and defect logs.

HARDWARE/SOFTWARE REQUIREMENTS

Hardware Requirements

- Processor: Intel i3 or equivalent
- RAM: 4-8 GB
- Hard Disk: 10 GB free space
- Internet Connection: Required for Jira and online resources

Software Requirements

- OS: Windows / Linux / macOS
- Browser: Google Chrome / Firefox
- Jira account (free cloud version)
- IDE/Editor: Visual Studio Code / Eclipse

Total(P)		60 Periods
Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy
CO1	Explain the concepts of automation testing, differentiate it from manual testing, and identify appropriate scenarios	L2 – Understand
CO2	Identify and apply various Selenium locators and WebDriver commands to interact with web elements	L3 - Apply
CO3	Demonstrate the ability to handle frames, alerts, windows, and perform advanced user interactions	L3 - Apply
CO4	Develop data-driven automated test scripts using Excel/flat files and implement synchronization strategies	L3 - Apply
CO5	Design and implement maintainable test automation frameworks using POM, Page Factory, and integrate logging	L4 - Analyze


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Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2							2
CO2	2	2	3	3					2
CO3	2	2	3	3			2		2
CO4	2	2	3	3	2				2
CO5	2	2	3	3	2	2	2		2
Average	2.2	2	3	3	2	2	2		2
1-Low, 2-Medium, 3-High									


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Vertical III : GENERAL VERTICAL

MC23CA425	EXPLORATORY DATA ANALYTICS AND VISUALIZATION	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To outline an overview of data analytics.					
2.	To perform univariate and bivariate data exploration analytics.					
3.	To use Data exploration and visualization techniques for multivariate and time series data.					
4.	To identify and classify various chart types for effective data visualization.					
5.	To explain the design components and standards for creating meaningful data visuals.					
	INTRODUCTION (Not for Examination)					2
Importance	The foundation of data science and visualization is crucial in today's data-driven world. It encompasses the core principles, methodologies, and tools required to transform raw data into actionable insights.					
Real-life Examples	- Used in healthcare to predict and visualize patient readmissions - Detecting fraudulent transactions in real-time - Transportation management in optimizing traffic flow and reducing congestion in urban areas - Retail inventory management to predicting stock levels and optimize inventory					
Linkages	Pre-requisite: Programming in Python, Mathematics for business analytics. Future Courses: Machine Learning, Big Data Analytics, Deep Learning					
UNIT-I	EXPLORATORY DATA ANALYTICS					9
	Exploratory data analytics - Understanding data science - Significance of EDA - Making sense of data - Comparing EDA with classical and Bayesian analysis - Data transformation techniques-merging database, reshaping and pivoting, Transformation techniques - Data Analytics - Managing Data for Analytics - Types of Data Analytics - Popular Data Analytics Methods - Data Visualization and Data Analytics. ❖ Project Topic: Customer Behaviour Analysis Using E-Commerce Data Analyze an e-commerce dataset (like user clicks, purchases, or product views) to understand customer behavior.					


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UNIT-II	UNIVARIATE AND BIVARIATE ANALYSIS	9
	Introduction to Single variable: Distributions and Variables - Numerical Summaries of Level and Spread - Scaling and Standardizing - Inequality - Relationships between Two Variables - Percentage Tables - Analyzing Contingency Tables - Handling Several Batches - Scatterplots and Resistant Lines. ❖ Project Topic: Analyzing Student Academic Performance Examine how different variables (study time, attendance, internet usage) affect student scores.	
UNIT-III	MULTIVARIATE AND TIME SERIES ANALYSIS	9
	Introducing a Third Variable - Causal Explanations - Three- Variable Contingency Tables and Beyond - Longitudinal Data - Fundamentals of TSA - Characteristics of time series data - Data Cleaning - Time-based indexing - Visualizing - Grouping - Resampling. ❖ Project Topic: Sales Trend Analysis for a Retail Chain Investigate multivariate influences and time-based sales trends.	
UNIT-IV	DATA VISUALIZATIONS	9
	Data visualization: Principles of Data visualization - Chart Types: Comparing Categories - Time - Distribution - Geospatial - Relationship - Part to whole - Qualitative - Tables. ❖ Project Topic: Covid-19 Global Data Dashboard Create an interactive dashboard to visualize pandemic trends.	
UNIT-V	DASHBOARD DESIGN	9
	Dashboard - Categorizing Dashboards - Power of Visual Perception: Limits of Short - Term Memory - Visually Encoding Data for Rapid Perception - Encoding Quantitative Versus Categorical Data - Gestalt Principles of Visual Perception - Effective Dashboard Display Media. ❖ Project Topic: Redesigning Ineffective Government Data Visuals Identify poorly designed visuals from government websites and improve them.	
	Total(L)	47 Periods
	❖ Internal Evaluation Only	


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OPEN-ENDED PROBLEMS / QUESTIONS			
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.		
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy	
CO1	Understand the fundamentals of data analytics.	L3 - Apply	
CO2	Perform univariate & bivariate data exploration and analytics.	L3 - Apply	
CO3	Use Data exploration and visualization techniques for multivariate and time series data.	L3 - Apply	
CO4	Differentiate and apply appropriate visualization techniques based on data types and objectives.	L3 - Apply	
CO5	Identify and describe effective dashboard display media and their suitability for different types of data and users.	L2 - Understand	
TEXT BOOKS			
1.	Suresh Kumar Mukhiya, Usman Ahmed, "Hands-On Exploratory Data Analysis with Python", Packt Publishing, 2020.		
2.	Jonathan Schwabish, "Better data visualizations: a guide for scholars, researchers, and wonks", 2 nd Edition, Columbia university Press, 2020.		
REFERENCE BOOKS			
1.	Stephen Few, "Information Dashboard Design", 1 st Edition, Oreilly, 2006.		
2.	Jake Vander Plas, "Python Data Science Handbook: Essential Tools for Working with Data", 1 st Edition, Oreilly, 2016.		
3.	Eric Pimpler, "Data Visualization and Exploration with R", GeoSpatial Training service, 2017.		
4.	Claus O. Wilke, "Fundamentals of Data Visualization", O'reilly publications, 2019.		
5.	A. G. W. Steyn, Rolf Stumpf, S. H. C. Du Toit, "Graphical Exploratory Data Analysis", Springer, 2012.		
WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content
1.	Towards Data Science Society	https://towardsdatascience.com/data-exploration-understanding-and-visualization-72657f5eac41	Articles
2.	Tableau	https://www.tableau.com/learn/articles/data-visualization	Articles
3.	Github reader	https://mschermann.github.io/data_viz_reader/introduction.html	Articles

VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Data Science Dojo	Lecture	https://www.youtube.com/watch?v=oskDpBfxBEI
2.	YouTube	Digital Humanities	Lecture	https://www.youtube.com/watch?v=b0R8vpxMShU
3.	NPTEL	Shankar Narasimhan, Ragunathan Rengasamy	Lecture	https://nptel.ac.in/courses/106106179

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2							2
CO2	2	3	3						2
CO3				3					
CO4	2	2	3	3	2				2
CO5	2	2	3	3	3	2			2
Average	2.2	2.2	3	2.8	2.5	2			2
1-Low, 2-Medium, 3-High									




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MC23CA426	CLOUD SERVICES MANAGEMENT	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand the fundamentals of cloud service models, deployment, and service management.					
2.	To gain knowledge on strategies for cloud service adoption, policy, and change management.					
3.	To learn service lifecycle, benchmarking, and migration techniques in cloud environments.					
4.	To explore pricing models and cost-related strategies for cloud services.					
5.	To comprehend governance, risk management, and value measurement of cloud services.					
	INTRODUCTION (Not for Examination)					2
Importance	It provides a comprehensive understanding of cloud service management by covering foundational concepts, strategic planning, operational management, economic models, and governance frameworks. It equips learners with the ability to design, adopt, manage, and evaluate cloud services effectively. By integrating technical, business, and security aspects, the course prepares students to make informed decisions in cloud-based environments. This knowledge is essential for roles in cloud architecture, DevOps, IT governance, and cloud financial management.					
Real-life Example(s)	Online storage platforms, video streaming services and collaborative productivity tools.					
Linkages	Pre-requisite: Computer Networks, Operating Systems.					
UNIT-I	CLOUD SERVICE MANAGEMENT FUNDAMENTALS					9
	Cloud Ecosystem – The Essential Characteristics – Basics of Information Technology Service Management and Cloud Service Management – Service Perspectives – Cloud Service Models – Cloud Service Deployment Models.					
UNIT-II	CLOUD SERVICES STRATEGY					9
	Cloud Strategy Fundamentals – Cloud Strategy Management Framework – Cloud Policy – Key Driver for Adoption – Risk Management – IT Capacity and Utilization – Demand and Capacity matching – Demand Queueing – Change Management – Cloud Service Architecture.					
UNIT-III	CLOUD SERVICE MANAGEMENT					9
	Cloud Service Reference Model – Cloud Service LifeCycle – Basics of Cloud Service Design – Dealing with Legacy Systems and Services – Benchmarking of Cloud Services – Cloud Service Capacity Planning – Cloud Service Deployment and Migration – Cloud Marketplace – Cloud Service Operations Management.					

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UNIT-IV	CLOUD SERVICE ECONOMICS	9
	Pricing models for Cloud Services – Freemium – Pay Per Reservation – Pay per User – Subscription based Charging – Procurement of Cloud-based Services – Capex vs Opex Shift – Cloud service Charging – Cloud Cost Models.	
UNIT-V	CLOUD SERVICE GOVERNANCE & VALUE	9
	IT Governance Definition – Cloud Governance Definition – Cloud Governance Framework – Cloud Governance Structure – Cloud Governance Considerations – Cloud Service Model Risk Matrix Cloud Security Fundamentals – Shared Responsibility Model – Identity and Access Management (IAM) – Cloud Data Protection (Encryption, Tokenization) – Network Security in Cloud (Firewalls, Zero Trust) – Security Monitoring and Incident Response – Compliance & Regulatory Standards (ISO 27001, SOC2, GDPR).	
	Total(L)	47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Describe cloud service models, deployment models, and basic concepts of service management.	L2 - Understand
CO2	Explain strategies for cloud adoption, risk, and capacity management	L2 - Understand
CO3	Explain cloud service lifecycle and analyze management and migration of cloud services.	L3 - Apply
CO4	Compare different cloud pricing models and explain the economic considerations.	L3 - Apply
CO5	Describe cloud governance structures and measure the value of cloud services.	L3 - Apply
	TEXTBOOKS	
1.	Thomas Erl, Zaigham Mahmood, Ricardo Puttini, "Cloud Computing - Concepts, Technology and Architecture", 1 st Edition, Prentice Hall, 2013.	
2.	Leo Chao, "Cloud Computing Networking: Theory, Practice, and Development", 1 st Edition, Auerbach Publications, 2015.	
3.	Tim Mather, Subra Kumaraswamy, Shahed Latif, "Cloud Security and Privacy", 1st Edition, O'Reilly, 2009.	


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REFERENCE BOOKS				
1.	Rajkumar Buyya, James Broberg, Andrzej Goscinski, "Cloud Computing: Principles and Paradigms", 1 st Edition, Wiley, 2013.			
2.	Barrie Sosinsky, "Cloud Computing Bible", 1 st Edition, O.Reilly, 2011.			
3.	Michael J. Kavis, "Architecting the Cloud", 1 st Edition, Wiley, 2014.			
4.	Ronald L. Krutz, Russell Dean Vines, "Cloud Security: A Comprehensive Guide", Wiley, 2010.			
WEB REFERENCES				
S. No.	Publisher	Website link	Type of Content	
1.	aws	https://aws.amazon.com/pricing/?ams%23interactive-card-vertical%23pattern-data.filter=%257B%2522filters%2522%253A%255B%255D%257D	Learning Content	
2.	Microsoft	https://azure.microsoft.com/en-us/products/cost-management/	Learning Content	
VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	David Lurie	Lecture	https://www.youtube.com/watch?v=kUNBx1MTwxw
2.	YouTube	Neudesic	Lecture	https://www.youtube.com/watch?v=urFy-6tO8PQ

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2		2					2
CO2	2	3		2	2				2
CO3	2	2	3	2	2				2
CO4	2	2	3	2	3				2
CO5	2	2	3	2	3	2			2
Average	2.2	2.2	3	2	2.5	2			2
1-Low, 2-Medium, 3-High									


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MC23CA431	EXPLORATORY DATA ANALYTICS AND VISUALIZATION LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand fundamental principles of data visualization and graphical representation.					
2.	To apply techniques to construct basic charts and plots for univariate and bivariate data.					
3.	To analyze distribution patterns and correlations using advanced charts and plots.					
4.	To implement interactive and map-based visualizations using tools like Plotly or PowerBI.					
5.	To design dashboards effectively considering user experience and principles of visual perception					
LIST OF EXPERIMENTS						
1. Introduction to Python for Data Analytics Write a Python program to perform basic data handling operations using Python libraries such as NumPy and Pandas. Demonstrate data creation, indexing, slicing, and basic statistical functions.						
2. Data Loading and Inspection Load a dataset from CSV/Excel format using Pandas and perform initial data inspection using functions like head, tail, info, describe, and shape to understand the structure and quality of data.						
3. Data Cleaning and Preprocessing Perform data cleaning operations on a real-world dataset by handling missing values, removing duplicates, correcting data types, and detecting outliers using Pandas and NumPy.						
4. Exploratory Data Analysis (EDA) Conduct exploratory data analysis using statistical techniques to identify patterns, trends, and relationships in the dataset. Compute measures such as mean, median, mode, variance, and correlation.						
5. Data Visualization using Matplotlib Create basic data visualizations using Matplotlib including line charts, bar charts, pie charts, and histograms to represent data distributions and trends effectively.						
6. Advanced Visualization using Seaborn Develop advanced visualizations using Seaborn such as box plots, violin plots, heatmaps, and pair plots to analyze relationships between multiple variables.						
7. Time Series Data Analysis Analyze time series data by visualizing trends, seasonality, and patterns over time. Plot time-based data and perform rolling statistics to observe changes across periods.						
8. Categorical Data Analysis Analyze categorical data using count plots, bar plots, and cross-tabulation. Interpret frequency distributions and relationships between categorical variables.						
9. Interactive Data Visualization Create interactive visualizations using tools such as Plotly or Dash to allow users to explore data dynamically through zooming, filtering, and hovering.						
10. Mini Project: Exploratory Data Analytics & Visualization Develop a mini project by selecting a real-world dataset (healthcare, finance, education, retail, etc.) and performing: a. Data collection b. Data preprocessing c. Exploratory data analysis						


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- d. Meaningful visualizations
- e. Insight generation
- f. Present findings using charts and summary reports.

HARDWARE/SOFTWARE REQUIREMENTS

Hardware Requirements

- Processor: Intel i3 or equivalent
- RAM: 4–8 GB
- Hard Disk: 10 GB free space
- Internet connection for accessing datasets and visualization tools

Software Requirements

- OS: Windows / Linux / macOS
- IDE/Editor: Jupyter Notebook / Visual Studio Code / R Studio
- Python (latest stable version) with libraries: Matplotlib, Seaborn, Plotly, Pandas
- PowerBI Desktop (optional, for interactive dashboards)
- Web Browser: Google Chrome / Firefox (latest version)

		Total(P)	60 Periods
Course Outcomes: Upon completion of this course, the students will be able to:			BLOOM'S Taxonomy
CO1	Construct bar charts and column charts using sample datasets		L3 - Apply
CO2	Use line and area charts to depict trends over time		L3 - Apply
CO3	Generate histograms, box plots, and violin plots to analyze distribution patterns		L3 - Apply
CO4	Create choropleth and dot maps using Plotly or PowerBI		L3 - Apply
CO5	Evaluate Dashboard Design Using Principles of Visual Perception		L4 - Analyze

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2	3	3					2
CO2	2	2	3	3					2
CO3	2	2	3	3					2
CO4	2	2	3	3			2		2
CO5	2	2	3	3	2	2	2		2
Average	2.2	2	3	3	2	2	2		2
1-Low, 2-Medium, 3-High									


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MC23CA432	CLOUD SERVICES MANAGEMENT LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.C.A - Master of Computer Applications	Version: 1.0				
Course Objectives						
1.	To understand cloud infrastructure concepts and implement role-based access control in cloud organizations.					
2.	To analyze cloud service usage and design cost models for applications.					
3.	To apply monitoring and alerting techniques to manage cloud resources effectively.					
4.	To evaluate billing and usage reports to optimize cloud costs.					
5.	To compare cloud services and make informed decisions on cloud provider selection based on cost-benefit analysis.					
LIST OF EXPERIMENTS						
1.	Study of Cloud Computing Basics Cloud concepts, service models (IaaS, PaaS, SaaS) and deployment models					
2.	Creating and Managing Cloud Account Create a cloud account and explore dashboard, regions, and billing					
3.	Virtual Machine Creation and Management Launch, start, stop, reboot, and terminate a virtual machine					
4.	Cloud Storage Management Create cloud storage and upload, download, and delete files					
5.	Identity and Access Management (IAM) Create users, roles, and assign permissions					
6.	Web Application Deployment on Cloud Deploy and host a simple web application					
7.	Cloud Database Services Create a cloud database and perform basic CRUD operations					
8.	Monitoring Cloud Resources Monitor VM performance and resource usage using cloud tools					
9.	Cloud Security and Backup Configure security groups, backups, and snapshots					
10.	Cloud Cost Management Analyze resource usage and billing reports					
HARDWARE/SOFTWARE REQUIREMENTS						
Hardware Requirements						
- Processor: Intel i3 or equivalent - RAM: 4-8 GB - Hard Disk: 10 GB free space - Internet Connection: Required for accessing cloud services						
Software Requirements						
- OS: Windows / Linux / macOS - Web Browser: Google Chrome / Firefox (latest versions) - Cloud Accounts: AWS, Azure, Google Cloud (free tiers sufficient) - Optional Open Source Cloud Platforms: OpenStack, Eucalyptus, OpenNebula - Spreadsheet/Editor: Excel / Google Sheets for cost modeling						
					Total(P)	60 Periods

w.e.f. 22.12.2025
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Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy
CO1	Explain fundamental cloud computing concepts, service models, deployment models, and cloud infrastructure basics.	L2 - Understand
CO2	Create a Cloud Organization with role-based access control (RBAC) by defining users, roles, and permissions.	L3 - Apply
CO3	Develop a cost model for a cloud-based web application and perform cost-benefit analysis.	L3 - Apply
CO4	Implement alerts for cloud resource usage to monitor and control performance.	L3 - Apply
CO5	Set up and analyze billing alerts for a Cloud Organization to optimize cloud expenditure.	L3 - Apply

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2	3	3					2
CO2	2	2	3	3	3				2
CO3	2	2	3	3					2
CO4	2	2	3	3	3				2
CO5	2	2	3	3	3				2
Average	2.2	2	3	3	2.6				2

1-Low, 2-Medium, 3-High

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MC23MB501	PRINCIPLES OF MANAGEMENT		CP	L	T	P	C
			3	3	0	0	3
Types of Courses	Open Elective						
Offering Dept.	MBA						
Programme & Branch	M.C.A - Master of Computer Applications					Version: 1.0	
Course Objectives							
1	To understand the nature, scope and functions of management.						
2	To explain various approaches and schools of management thought.						
3	To analyse the functional areas, skills and roles required for effective management.						
4	To apply planning and decision-making techniques in managerial situations.						
5	To evaluate modern concepts and the importance of management in organizational success.						
UNIT-I	NATURE AND FUNCTIONS OF MANAGEMENT					9	
	Definition and Nature of Management – Features of Management – Functions of Management – Management as a Process – Importance of Management – Management vs. Administration.						
UNIT-II	HISTORY OF MANAGEMENT					9	
	Schools of Management Thought – Classical Approaches: Systematic Management, Scientific Management, Administrative Management, Human Relations Movement – Contemporary Approaches: Quantitative Management, Organizational Behaviour, Systems Theory, Contingency Theory.						
UNIT- III	BASICS OF MANAGEMENT					9	
	Functional Areas of Management – Managerial Skills – Roles of a Manager – Levels of Management – Management as a Science, an Art, and a Profession.						
UNIT – IV	PLANNING					9	
	Planning – Meaning and Definition – Features of Planning – Steps in Planning Process – Approaches to Planning – Principles of Planning – Importance of Planning – Advantages and Disadvantages of Planning – Types of Plans – Types of Planning – Management by Objectives (MBO).						
UNIT-V	DECISION-MAKING					9	
	Decision-Making – Meaning – Characteristics of Decision- Making – Decision-Making Process – Guidelines for Effective Decision Making – Types of Decisions.						
	TOTAL(L)					45 PERIODS	

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	OPEN ENDED PROBLEMS / QUESTIONS	
1.	Analyze how managerial functions influence organizational success in modern companies.	
2.	Compare classical and contemporary management approaches with real-world examples.	
3.	Develop a planning model for a start-up and justify each step in the planning process.	
4.	Evaluate the impact of decision-making styles on team effectiveness.	
5.	Prepare a case study on how management as a profession is emerging in India.	
	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
CO1	Recall the basic concepts, nature, functions, and significance of management.	L3 - Apply
CO2	Identify the functional areas of management, along with the roles and skills of managers.	L3 - Apply
CO3	Recognize the major classical and contemporary schools of management thought.	L3 - Apply
CO4	Explain the planning process, principles, and various types of plans including MBO.	L3 - Apply
CO5	Describe the stages of managerial decision-making and various types of decisions.	L3 - Apply
	TEXT BOOKS	
1.	Harold Koontz & Heinz Weihrich, Essentials of Management: An International, Innovation and Leadership Perspective, McGraw-Hill Education, Tenth Edition, 2015.	
2.	Stephen P. Robbins & Mary Coulter, Management, Pearson Education, Thirteenth Edition, 2017.	
	REFERENCE BOOKS	
1.	Ramaswamy T., Principles of Management, Himalaya Publishing House, First Edition.	
2.	L. M. Prasad, Principles and Practices of Management, Sultan Chand & Sons, Eighth Edition, 2015.	
3.	Peter F. Drucker, The Practice of Management, Harper & Row Publishers, First Edition, 1954.	
4.	Stoner, J. A. F., Freeman, R. E. & Gilbert, D. R., Management, Pearson Education, Sixth Edition, 2006.	


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WEB REFERENCES				
S. No.	Publisher	Website link		Type of Content
1.	Open Textbook Librar	https://open.umn.edu/opentextbooks		Management concepts, planning, decision-making (free textbooks)
2.	Investopedia	https://www.investopedia.com		Management basics, decision making, MBO, managerial roles
VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	Introduction to Management & Functions	Management & Functions Prof. Anil Kumar Neerukonda (NPTEL)	Conceptual Lecture	https://nptel.ac.in/courses/110/105/110105035
2.	Planning and Decision Making	Prof. R. Srinivasan (IIT Madras – NPTEL)	Academic Lecture	https://nptel.ac.in/courses/110/107/110107142

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	1							2
CO2	3	2							2
CO3	2	2							2
CO4	2	3	2						2
CO5	2	4	2						3
Average	2.4	2.2	2						2.2
1-Low, 2-Medium, 3-High									


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MC23MB502	ORGANIZATIONAL BEHAVIOUR		CP	L	T	P	C
			3	3	0	0	3
Types of Courses	Open Elective						
Offering Dept.	MBA						
Programme & Branch	M.C.A - Master of Computer Applications					Version: 1.0	
Course Objectives							
1	To understand the need and importance of the organizational behavior.						
2	To explain the various personality and perception.						
3	To recognize the concepts of group behaviour and communication.						
4	To discuss concept of Leadership and Emotional Intelligence.						
5	To discuss the dynamics of organizational behaviour.						
UNIT-I	FOCUS AND PURPOSE					9	
	Definition, need and importance of organizational behaviour – Nature and scope – Frame work – Organizational behaviour model.						
UNIT-II	INDIVIDUAL BEHAVIOUR					9	
	Personality – types – Factors influencing personality – Theories – Learning – Types of learners – The learning process – Learning theories – Organizational behaviour modification. Misbehaviour – Types – Motivation – Importance – Types – theories - Effects on work behaviour.						
UNIT- III	GROUP BEHAVIOUR					9	
	Group Formation – Groups in organizations – Influence – Group dynamics – Emergence of informal leaders and working norms – Group decision making techniques – Team building - Interpersonal relations – Communication – Control.						
UNIT – IV	LEADERSHIP, POWER AND EMOTIONAL INTELLIGENCE					9	
	Meaning – Importance – Leadership styles – Theories – Leaders Vs Managers – Sources of power – Power centers – Power and Politics. Emotions - Emotional Labour – Emotional Intelligence – Theories.						
UNIT-V	DYNAMICS OF ORGANIZATIONAL BEHAVIOUR					9	
	Organizational culture and climate – Factors affecting organizational climate – Importance. - Job satisfaction – Determinants – Measurements – Influence on behaviour. Organizational change – Importance – Stability Vs Change – Proactive Vs Reaction change – the change process.						
	Total(L)					45 PERIODS	

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	OPEN ENDED PROBLEMS / QUESTIONS		
1.	Study about behaviour model influence in industry.		
2.	Analysis and categories the personality of employees.		
3.	Study the perception of employees in nearest companies.		
4.	Study the emotional intelligence of employees.		
5.	Study the stress coping strategies followed by employees.		
	Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Understand the basic concepts of managing Organizational Behaviour.		L3 - Apply
CO2	Demonstrate the Individual Differences, perception, learning and motivation		L3 - Apply
CO3	Apply the concepts of Group dynamics, Interpersonal relations and communication.		L2 - Understand
CO4	Describe the concepts of leadership, emotional intelligence and attitudes.		L3 - Apply
CO5	Understanding on work stress and work life balance.		L3 - Apply
	TEXT BOOKS		
1.	Stephen P. Robbins, Timothy A. Judge, Organisational Behaviour, PHI Learning / Pearson Education, 16th edition, 2014.		
2.	K. Aswattappa, Organisational Behaviour, Himalaya Publishing House, 12th Edition, 2016.		
	REFERENCE BOOKS		
1.	Prasad .L.M., Organisational Behaviour ,Sultan Chand and Sons, 2019		
2.	C.B.Guptha, A Textbook Of Organisational Behaviours ,S.Chand & Company,2019		
3.	Luthans, F. Organizational Behaviour, 12th Edition, Tata McGraw Hill Education, 2017.		
4.	McShane, S.L., Von Glinow, M.A., and Sharma, R.R., Organizational Behaviour, 5th Edition, Tata McGraw-Hill Education Pvt. Ltd., 2011		
	WEB REFERENCES		
S. No.	Publisher	Website link	Type of Content
1.	Open Textbook Library	https://open.umn.edu/opentextbooks	Learning Content
2.	Investopedia	https://www.investopedia.com	Learning Content

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VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	David Lurie	Lecture	https://www.youtube.com/watch?v=kUNBx1MTwxw
2.	YouTube	Neudesic	Lecture	https://www.youtube.com/watch?v=urFy-6tO8PQ
3.	YouTube	NPTEL (IIT Faculty)	Lecture	https://nptel.ac.in/courses/110/105/110105038

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	1							2
CO2	3	2							2
CO3	2	2				2	3		2
CO4	2	2				2	2	2	2
CO5	2	3				1	2	2	3
Average	2.4	2				1.4	1.8	0.8	2.2

1-Low, 2-Medium, 3-High


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MC23MB503	FUNDAMENTALS OF MARKETING				CP	L	T	P	C
					3	3	0	0	3
Types of Courses	Open Elective								
Offering Dept.	MBA								
Programme & Branch	M.C.A - Master of Computer Applications						Version: 1.0		
Course Objectives									
1	To understand the basic concepts of marketing.								
2	To examine the consumer buying behavior and its process.								
3	To understand the marketing mix of various products and the concept of retailing, wholesaling and Logistics.								
4	To determine the pricing strategies followed by different business organizations.								
5	To understand new concepts and techniques of market research.								
UNIT-I	INTRODUCTION TO MARKETING						9		
	Core concepts - Evolution of marketing - Prospects and challenges - Marketing interface with other functional areas.								
UNIT-II	CONSUMER BEHAVIOUR						9		
	Influencing factors - Customer buying process - Customer acquisition, satisfaction, loyalty - Customer relationship management.								
UNIT-III	MARKETING MIX						9		
	Product, price, place, promotion - Product life cycle - Retailing, wholesaling and logistics.								
UNIT-IV	DESIGNING PRICING STRATEGIES						9		
	Pricing strategy - Types of Pricing - How to create a pricing strategy - Pricing models based on Industry or Business - Conducting a pricing analysis.								
UNIT-V	MARKET RESEARCH TECHNIQUES						9		
	Define market research - Importance of market research - Major categories of market research - Techniques of market research - Steps for conducting market research.								
	Total(L)						45 PERIODS		
	OPEN ENDED PROBLEMS / QUESTIONS								
	Course specific Open Ended Problems will be solved during the class room teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.								

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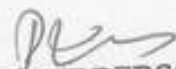
	Course Outcomes: Upon completion of this course the students will be able to:	Bloom's Taxonomy		
CO1	Understanding the modern marketing concepts in various functional areas.	L3 - Apply		
CO2	Analyzing the nature of consumer buying behavior.	L2 - Understand		
CO3	Applying the marketing mix of products.	L2 - Understand		
CO4	Implementing various pricing strategies in business organization.	L3 - Apply		
CO5	Applying the marketing research techniques and new trends in the field of Marketing.	L3 - Apply		
	TEXT BOOKS			
1	Kotler.P, Keller K, Koshy, Jha (2012), Marketing Management, A south Asian Perspective, 14 th edition, Pearson Education.			
2	Philip Kotler (2002), Framework for Marketing Management, New Delhi, Person Education.			
	REFERENCE BOOKS			
1	Philip Kotler, Kevin Lane Keller, Alexander Chervev, Jagdish N. Sheth G. Shainesh, Marketing Management (16th Edition) Pearson India Education service Pvt. Ltd			
2	Leon G. Schiffman, Leslie Iazar Kanuk, Ramesh Kumar (2010), Consumer Behaviour , 10 th Edition Pearson Education.			
3	RajendraNargundkar, Marketing Research : text and cases			
	WEB REFERENCES			
S. No.	Publisher	Website link	Type of Content	
1.	Kotler Marketing Group	https://www.kotlerimpact.com	Web Reference	
2.	Investopedia	https://www.investopedia.com	Web Reference	
3.	American Marketing Association	https://www.ama.org/	Web Reference	
4.	Tutorials Point	https://www.tutorialspoint.com/marketing management	Web Reference	
5.	HubSpot Academy	https://academy.hubspot.com	Web Reference	
6.	Nielsen	https://www.nielsen.com	Web Reference	
	VIDEO REFERENCES			
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Philip Kotler	Tutorial	https://www.youtube.com/watch?v=qVZ0xA
2.	YouTube	NPTEL (IIT Faculty)	Tutorial	https://nptel.ac.in/courses/110105/5034

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Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
C01	3	1							2
C02	2	3							2
C03	2	2	2						2
C04	2	2	3						2
C05	2	3	2						3
Average	2.2	2.2	1.4						2.2
1-Low, 2-Medium, 3-High									


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MC23MB504	ACCOUNTING FOR MANAGERS		CP	L	T	P	C
			3	3	0	0	3
Types of Courses	Open Elective						
Offering Dept.	MBA						
Programme & Branch	M.C.A - Master of Computer Applications					Version : 1.0	
Course Objectives							
1.	To enable students to understand accounting principles, processes, and policies used for preparing financial statements.						
2.	To acquire conceptual knowledge of the corporate accounting system and to learn the techniques of preparing the financial statements of companies.						
3.	To students study accounting tools and techniques so as to develop their skills for analyzing and presenting financial statements for making business decisions in the current environment.						
4.	To study the costing concept and methods.						
5.	To discuss the types of budget and illustrate their preparation.						
UNIT-I	FINANCIAL ACCOUNTING					9	
	Significance and importance of Accounting in business organizations, Branches of accounting: Financial Accounting, Cost Accounting, Management Accounting and contemporary developments (Human Resource Accounting, Brand Accounting, Inflation Accounting, Social Accounting, Environmental Accounting) Concept of GAAP Concept of Accounting equation & computation of profit from business using the same.						
UNIT-II	CORPORATE ACCOUNTING					9	
	Introduction to Corporate Accounting – Concept of Company, Concept of Share capital and types, Disclosure requirements, Contents of Annual Report.Basic Concepts in Tally Software.						
UNIT- III	ANALYSIS OF FINANCIAL STATEMENTS					9	
	Financial Statement Analysis – Ratio Analysis, Fund Flow Statement & Cash Flow Statement (Conceptual understanding alone).						
UNIT – IV	COST ACCOUNTING						
	Cost Accounting – Basic Definition, Elements of costs, Job Cost Sheet (Problems), Batch Costing, Contract Costing and Process Costing, (Single Process, without and with closing WIP) ABC & Target Costing.						
UNIT-V	BUDGETING AND VARIANCE ANALYSIS					9	
	Techniques of Cost Accounting – Marginal Costing, Make or buy Decisions, Standard Costing (Material and Labour variances only), Budget (Problems in Production and Cash), Budgeting and Budgetary Control Techniques.						
	TOTAL(L)					45 PERIODS	

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	OPEN ENDED PROBLEMS / QUESTIONS	
1.	A Study on Utility of Accounting System among MSME Units in Salem zone.	
2.	Forfeiture of Shares at a Premium.	
3.	The usefulness of financial ratio analysis as a tool for enhancement of corporate investment Decisions.	
4.	Cost accounting information and product costing in selected paint manufacturing companies.	
5.	Variance analysis as an effective tool for budgeting in an organization.	
	Course Outcomes: Upon completion of this course the students will be able to:	BLOOMS Taxonomy
CO1	To apply and analyze the financial statements and understand the role of IFRS/Ind-AS in the accounting discipline.	L2 - Understand
CO2	Acquire the knowledge of corporate accounts such as the meaning of a company, characteristics of a company's, types of shares and debentures.	L3 - Apply
CO3	Assess the financial soundness of the company using ratio analysis, cash flow and fund flow.	L5 - Evaluate
CO4	To analyze and evaluate the elements and classification of cost and overheads, levels of material control, purchase and store control.	L3 - Apply
CO5	To assess the forecasted behavior in finance by using budget and variance analysis.	L3 - Apply
	TEXT BOOKS	
1.	Dr. K.L. Gupta ,Accounting for Managerial Decisions, Sahitya Bhawan Publications, 2023	
2.	Maheswari S.N., CA Sharad K. Maheswari, Suneel K. Maheswari, "A Textbook of Accounting for Management", 4 th Edition, Vikas Publishing, New Delhi, 2018.	
	REFERENCE BOOKS	
1.	Ambrish Gupta, "Financial Accounting for Management", 6th Edition, Pearson Education, Noida, 2018.	
2.	N. Ramachandran Ram Kumar Kakani,"Financial Accounting for Management", New Delhi: Tata McGraw Hill, 2015.	
3.	Shukla M.C. Grewal T.S. , Gupta M.P ,Cost Accounting: Texts And Problems, S Chand and Company Ltd.	
4.	T.S. Reddy & Dr. A. Murthy Corporate Accounting, Margham Publications.	
5.	R. Narayanaswamy, "Financial Accounting - A Managerial Perspective", New Delhi: Prentice Hall India, 2016.	


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WEB REFERENCES				
S. No.	Publisher		Website link	Type of Content
1.	Investopedia		https://www.investopedia.com	Web Reference
2.	Accounting Tools		https://www.accountingtools.com	Web Reference
3.	ICAI (Institute of Chartered Accountants of India)		https://www.icai.org	Web Reference
VIDEO REFERENCES				
S. No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	NPTEL (IIT Faculty)	Tutorial	https://nptel.ac.in/courses/110/105/110105123
2.	YouTube	CA Rachana Ranade	Tutorial	https://www.youtube.com/watch?v=ZK4w5E1h6jI

Mapping of COs with POs									
COs	POs								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2							2
CO2	3	1							2
CO3	2	3							2
CO4	2	2							2
CO5	2	3							3
Average	2.4	2.2							2.2
1-Low, 2-Medium, 3-High									


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